DIGITAL COMMUNICATIONS

Digital Communications

Lecture notes for the whole course: how a waveform becomes a finite alphabet of symbols, how those symbols are carried across a channel that adds noise to everything, how a receiver decides which one was sent, and how often it is wrong.

Covers	Level	Assumed background
Chapters 1 to 6, and appendix A	Undergraduate	Fourier analysis, probability, random processes

How to read these notes

Every chapter builds on the one before it. Within a chapter, each idea arrives in the same order: a picture, a definition, an equation, a short derivation, a worked example, and a warning about the mistake that is easiest to make. Worked examples use five headings — Given, Find, Method, Solution, Check. Do the Check step yourself before reading it.

Two conventions apply everywhere and both are worth fixing now, because getting either wrong costs a factor of two that nothing on the page reveals. Noise is white and Gaussian with **two-sided** power spectral density $N_0/2$ watts per hertz. The Gaussian tail is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2}\, dt = \frac{1}{2}\,\mathrm{erfc}(x/\sqrt{2})$. Energy and power are normalised, with $R = 1\ \Omega$.

The contents below carries a third column. An entry such as **PS CH7.2.1** points into the course textbook, Proakis and Salehi, *Fundamentals of Communication Systems*, second edition, where the same material is developed at length. The **PS** marker is what tells the two apart, and it is not decorative: these notes reach information theory in chapter 6 and the textbook develops it in chapter 12, while the textbook's own chapter 12 heading is nowhere near what a bare number would suggest.

CHAPTER 1

The transition from analog to digital

A continuous waveform becomes a bit stream in three steps: sampling makes it discrete in time, quantization makes it discrete in amplitude, and encoding replaces each amplitude by a word of bits. Only the first of the three can be undone, and this chapter is largely about what that means.

1.1 Impulse-train sampling

Sampling every T_s seconds is multiplication by a train of impulses. Write $f_s = 1/T_s$ for the sampling frequency.

THE IDEAL SAMPLED SIGNAL

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$g_\delta(t) = g(t)\,p(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\delta(t - nT_s)$$

The second line uses the sampling property $g(t)\delta(t-t_0) = g(t_0)\delta(t-t_0)$, which turns the product under the sum into a number.

What makes this worth writing is that g_δ is a continuous-time signal, not a sequence, so it has a Fourier transform. A product in time is a convolution in frequency, and the transform of an impulse train is another impulse train:

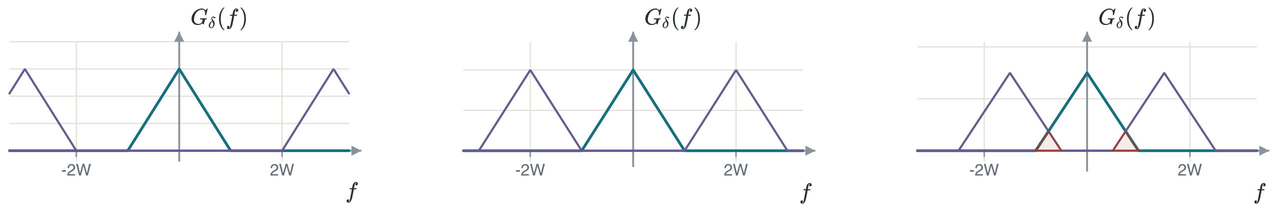
$$P(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

The coefficient is the Fourier-series coefficient of $p(t)$, and it is the same for every harmonic: $a_k = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi k f_0 t}\,dt = \frac{1}{T_s}$, by the sifting property. Convolving G with that train gives the result the rest of the chapter rests on.

SAMPLING REPLICATES THE SPECTRUM

$$G_\delta(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

Sampling copies the spectrum to every multiple of f_s and scales it by f_s. Nothing is lost provided the copies do not overlap.



$f_s > 2W$: a gap between the replicas. $f_s = 2W$: they touch and do not overlap. $f_s < 2W$: they overlap, and the overlap cannot be undone.

ALIASING

In the third case a high frequency of the message has been added to a low frequency of a replica. No filter can separate a sum into its parts, so the original signal cannot be recovered from its samples — not by a better filter and not by more computation. In practice a real signal is never strictly bandlimited, so a **guard band** f_g is left between the edge of the message and the edge of the first replica: $f_s = 2W + f_g$, with an anti-aliasing filter removing whatever lies above W before the sampler sees it.

1.2 The sampling theorem and reconstruction

SAMPLING THEOREM

If $G(f) = 0$ for $|f| \geq W$ and $f_s \geq 2W$, then $g(t)$ is determined completely by its samples $g(nT_s)$ and can be recovered from them exactly. If $f_s < 2W$, aliasing occurs and the recovery fails. The rate $2W$ is the **Nyquist rate** and $T_s = 1/(2W)$ the **Nyquist interval**.

Recovery means keeping the copy at the origin and rejecting every other one, which is a lowpass filter. At $f_s = 2W$:

THE RECONSTRUCTION FILTER AND ITS IMPULSE RESPONSE

$$H_{\mathrm{LPF}}(f) = \begin{cases} \dfrac{1}{2W}, & |f| \leq W \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\mathrm{LPF}}(t) = \int_{-W}^{W} \frac{1}{2W} e^{j2\pi ft}\, df = \frac{\sin(2\pi Wt)}{2\pi Wt} = \operatorname{sinc}(2Wt)$$

The convention used throughout is $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, so that sinc is one at zero and zero at every other integer.

THE GAIN IS NOT ONE

Sampling multiplied the spectrum by $f_s = 2W$, so the filter has to divide it back. A reconstruction filter of unit gain returns a signal $2W$ times too large, and nothing in a plot of the spectrum *shape* reveals it.

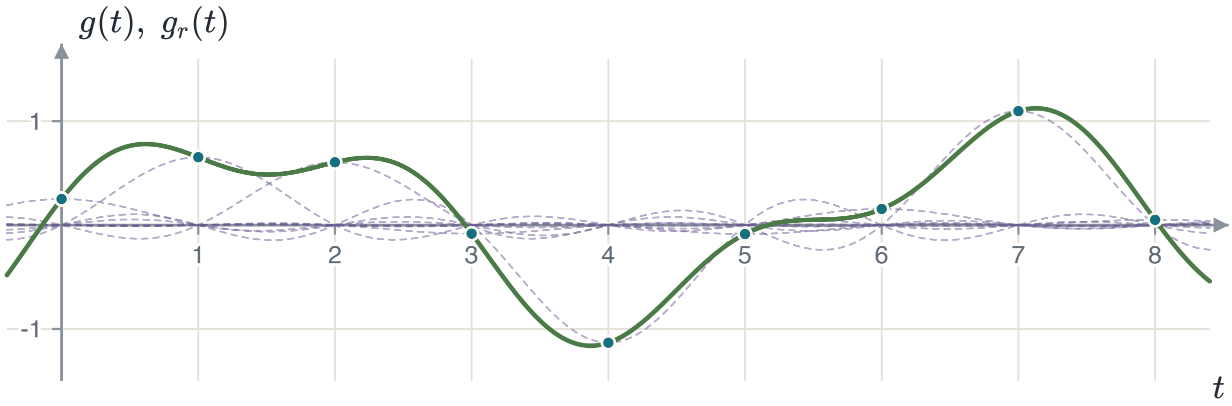
Filtering in frequency is convolution in time. Convolving the impulse train with h_{LPF} and using the sifting property once more gives the interpolation formula.

INTERPOLATION

$$g_r(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\mathrm{sinc}\big(2W(t-nT_s)\big)$$

$$g_r(t) = \sum_{n=-\infty}^{\infty} g\Big(\frac{n}{2W}\Big)\,\mathrm{sinc}(2Wt-n) \quad \text{at } f_s = 2W$$

At $t = kT_s$ every term vanishes except the one with $n = k$, because sinc is zero at every non-zero integer. So $g_r(kT_s) = g(kT_s)$ exactly — and, since the filter passes the whole message and nothing else, the two agree everywhere and not only at the instants.



Each sample contributes one shifted sinc scaled by its own value. Their sum, drawn heavy, passes through every sample because each sinc is zero at all the other sampling instants.

EXAMPLE 1.1 — THREE SAMPLING RATES

GIVEN A signal $x(t)$ bandlimited to $W = 40$ kHz.

FIND (a) the Nyquist rate; (b) the rate with a 10 kHz guard band; (c) the Nyquist rate of $y(t) = x(t)\cos(80000\pi t)$.

METHOD Read (a) and (b) off the geometry of the replicas. For (c) find the bandwidth of y first: the rate follows from that, not from the bandwidth of x.

SOLUTION (a) $f_s = 2W = 80$ kHz. (b) $f_s = 2W + f_g = 90$ kHz. (c) The carrier is at $f_c = 40$ kHz, and multiplication shifts the spectrum both ways and halves it: $Y(f) = \frac{1}{2}X(f-40\mathrm{k}) + \frac{1}{2}X(f+40\mathrm{k})$. The result occupies $|f| < 80$ kHz, so $f_s = 160$ kHz.

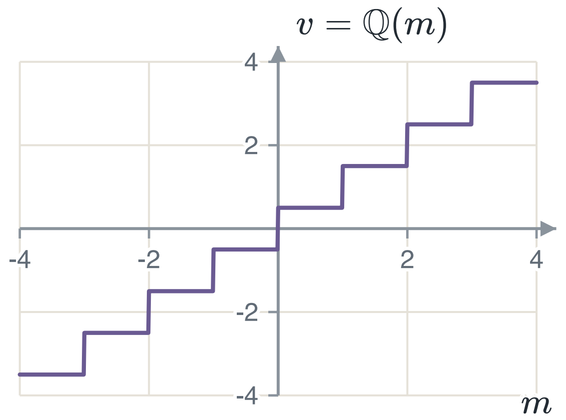
CHECK The highest frequency of y is $f_c + W = 80$ kHz and $2 \times 80 = 160$ kHz. Doubling the carrier would double the answer while leaving the width of Y unchanged: the rate follows the highest frequency present, not the width of the message.

THE TRAP IN PART (C)

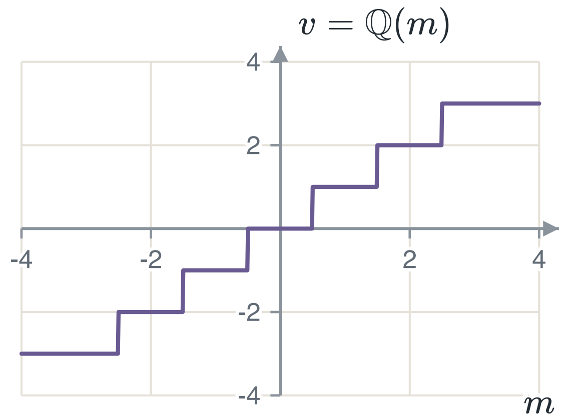
Answering 80 kHz is answering for x, not for y. Modulation moved the message up the frequency axis and the sampler has to keep up with where it went.

1.3 Quantization

Quantization replaces a sample amplitude by the nearest member of a finite set of L **representation levels**. In a **uniform** quantizer the spacing Δ between consecutive levels is the same everywhere. The two families differ in what happens at zero: a **mid-rise** quantizer puts a decision boundary there, so no output level is zero, while a **mid-tread** quantizer puts a level there, so a small input is quantized to exactly zero. Speech coders use mid-tread, because a silent input should be reproduced as silence rather than as a $\pm\Delta/2$ chatter.



Mid-rise, $L = 8$, $\Delta = 1$. A boundary at the origin.



Mid-tread, $L = 8$, $\Delta = 1$. A level at the origin.

A quantizer is a partition of the input range into L regions $\mathcal{J}_k$ with one output value v_k per region. Two conditions make it optimal, and they refer to each other, which is why they are applied by turns rather than solved at once.

1. Each boundary is the **midpoint** of the two levels it separates, $m_k = \frac{1}{2}(v_{k-1} + v_k)$. Given the levels, this minimises the error, because it sends every input to the nearer level.
2. Each level is the **centroid** of its own region, $v_k = E[M \mid M \in \mathcal{J}_k]$. Given the boundaries, this minimises the mean square error inside the region.

A uniform quantizer satisfies the first by construction. It satisfies the second only when the input is uniformly distributed, which is the deeper reason the uniform quantizer is optimal for a uniform source and for no other.

1.4 Quantization noise and the signal-to-noise ratio

The quantization error is $Q = M - \mathbb{Q}(M)$. Because every input is sent to the nearer level, it can never exceed half a step, so $-\Delta/2 \le q \le \Delta/2$. If Δ is small enough that the density of M is nearly flat across one region, the error is as likely to fall anywhere in that interval as anywhere else.

THE UNIFORM ERROR MODEL

$$f_Q(q) = \frac{1}{\Delta}, \qquad -\frac{\Delta}{2} \le q \le \frac{\Delta}{2}$$

$$E[Q^2] = \int_{-\Delta/2}^{\Delta/2} q^2 \frac{1}{\Delta}\, dq = \frac{\Delta^2}{12}$$

With a zero-mean input the error has zero mean too, so this mean square is also the variance: the power of the quantization noise is its variance.

Substituting $\Delta = 2m_{\max}/L$ and $L = 2^R$ puts the result in the form the rest of the chapter uses, and dividing the signal power by it gives the figure of merit.

SIGNAL-TO-QUANTIZATION-NOISE RATIO

$$E[Q^2] = \frac{1}{12}\left(\frac{2m_{\max}}{L}\right)^2 = \frac{m_{\max}^2}{3 \cdot 2^{2R}}$$

$$\mathrm{SQNR} = \frac{P_M}{E[Q^2]} = \frac{3P_M}{m_{\max}^2} 2^{2R}$$

$$\mathrm{SQNR}\ [\mathrm{dB}] = \underbrace{10 \log_{10} \frac{3P_M}{m_{\max}^2}}_{\alpha} +\ 6.02R$$

Every extra bit per sample adds $20\log_{10} 2 = 6.02$ dB. Doubling the level count halves the step, which quarters the error power, and a factor of four is 6.02 dB.

WHAT α COSTS

The intercept is negative for every signal that does not fill the quantizer range. A full-scale sinusoid has $P_M = m_{\max}^2/2$ and $\alpha = 10\log_{10} 1.5 \approx 1.76$ dB; a sinusoid reaching a quarter of the range has $\alpha = -10.28$ dB, a loss of 12.04 dB — exactly two bits. That calculation is the argument for companding, and its size is the size of the argument.

EXAMPLE 1.2 – A FULL-SCALE SINUSOID

GIVEN $m(t) = 5\cos t$ through a uniform quantizer spanning its full range.

FIND The step size and the SQNR at $R = 3$ and $R = 4$ bits per sample.

METHOD Average power from Parseval, step size from $\Delta = 2m_{\max}/L$, then $\alpha + 6.02R$.

SOLUTION The coefficients are $a_{\pm 1} = 5/2$, so $P_M = 2(5/2)^2 = 12.5$ and $m_{\max} = 5$. At $R = 3$: $\Delta = 2(5)/8 = 1.25$ V and $\mathrm{SQNR} = 1.76 + 18.06 = 19.82$ dB. At $R = 4$: $\Delta = 0.625$ V and $\mathrm{SQNR} = 1.76 + 24.08 = 25.84$ dB.

CHECK The two differ by 6.02 dB, one bit. Independently, $E[Q^2] = \Delta^2/12 = 0.1302$ and $10\log_{10}(12.5/0.1302) = 19.82$ dB.

WHAT THE FORMULA DOES NOT KNOW

Quantize this waveform and measure the error it actually makes, and the answers are 19.09 dB and 25.31 dB — about 0.7 dB and 0.5 dB below the formula. The uniform model assumes the error is equally likely anywhere in half a step, and a sinusoid spends most of its time near its peaks, where the error is largest. The gap halves with every extra bit: 0.27 dB at $R = 6$ and 0.14 dB at $R = 8$. That is what "Δ small enough" means, and knowing the size of the gap is the difference between using a model and believing it.

EXAMPLE 1.3 — WHEN THE MODEL BREAKS

GIVEN A zero-mean stationary Gaussian source with $S_X(f) = 2$ for $|f| < 100$ Hz, sampled at the Nyquist rate, each sample through a five-level quantizer with outputs $-30, -10, 0, 10, 30$ and boundaries at $-40, -20, 20, 40$.

FIND The SQNR of the scheme.

METHOD The signal power is the area under the spectral density. The noise power must be integrated region by region, because this quantizer is neither uniform nor fine.

SOLUTION $P_X = \int_{-100}^{100} 2\, df = 400$, and since the mean is zero this is also σ_X^2. Splitting $P_Q = \int (x - \mathbb{Q}(x))^2 f_X(x)\, dx$ at the four boundaries gives $7.98 + 46.36 + 79.50 + 46.36 + 7.98 = 188.18$, so $\mathrm{SQNR} = 10 \log_{10}(400/188.18) = 3.27$ dB.

CHECK The central region alone contributes 79.50, and it holds the 68% of the mass with $|X| < 20$ and an error of up to 20. That is where a five-level quantizer spends its error, and it is why the answer is a few decibels rather than a few tens.

WHY $\Delta^2/12$ WOULD HAVE LIED

The quantizer is coarse and its outer regions are unbounded, so the error there is not bounded by half a step at all: a sample at $x = 120$ is quantized to 30 and the error is 90. Applying the uniform model with $\Delta = 20$ predicts $E[Q^2] = 33.3$ and an SQNR of 10.8 dB, three times too optimistic. The model is a small-step model, and this is what its failure looks like.

1.5 Non-uniform quantization

Speech spends most of its time at small amplitudes and only occasionally reaches its peak. A uniform quantizer gives the same absolute step to a whisper and to a shout, so the whisper is quantized far more coarsely in proportion to itself. Relaxing the requirement that the regions be of equal length allows the distortion to be reduced at the same level count — narrow regions where the density is high and wide ones where it is low, which is what the centroid condition asks for and what a uniform quantizer cannot give.

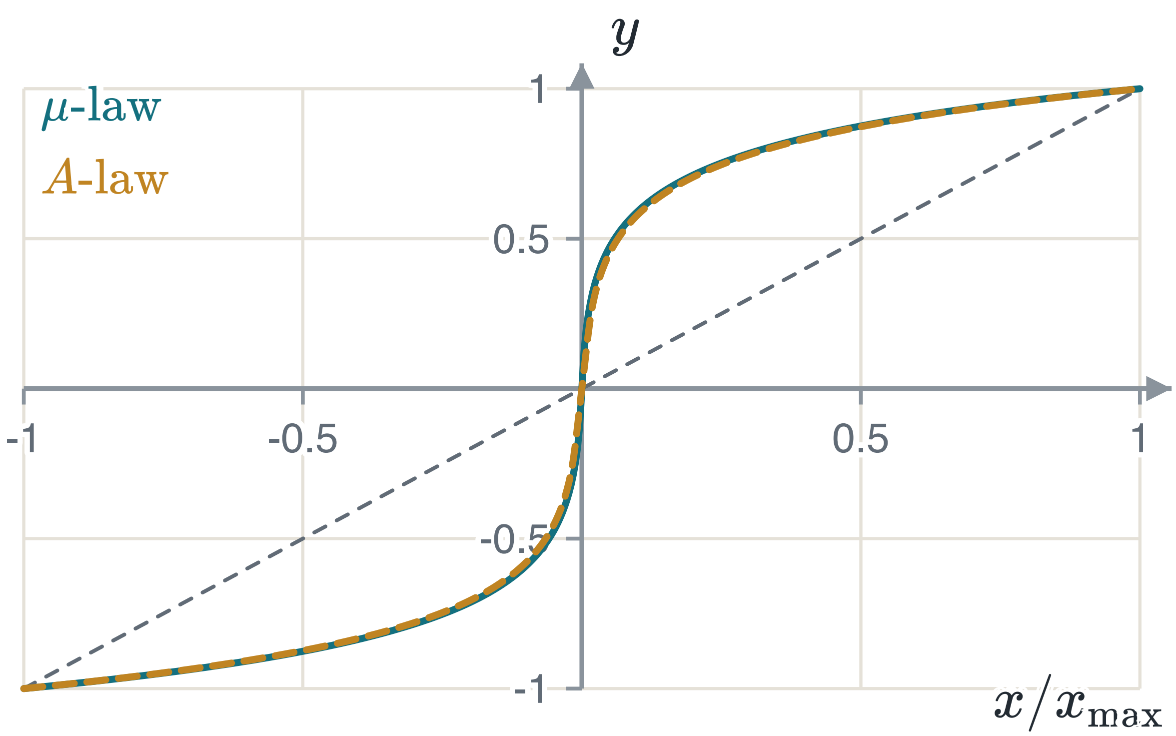
Rather than build a non-uniform quantizer, the signal is compressed by a memoryless non-linearity, quantized **uniformly**, and expanded at the receiver by the inverse. Compressing plus expanding is **companding**, and it is how the non-uniform quantizer is built out of a uniform one.

THE TWO STANDARD COMPRESSORS

$$y = \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}\,\mathrm{sgn}(x), \qquad |x| \le 1$$

$$y = \begin{cases} \dfrac{A|x|}{1+\ln A}\,\mathrm{sgn}(x), & 0 \le |x| \le 1/A \\ \dfrac{1+\ln(A|x|)}{1+\ln A}\,\mathrm{sgn}(x), & 1/A < |x| \le 1 \end{cases}$$

Both are normalised so that $x = \pm 1$ maps to $y = \pm 1$. The standard parameters are $\mu = 255$ and $A = 87.6$; on the same speech through the same eight-bit uniform quantizer the two give signal-to-noise ratios within a hundredth of a decibel of one another. The difference between them is regional rather than technical.

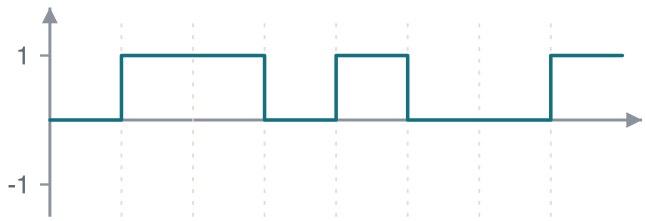


The two compressor characteristics against the identity, which is what no companding would give. Both spend most of their output range on the smallest tenth of the input.

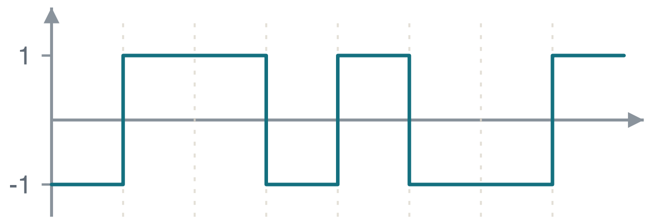
1.6 Encoding, line codes and pulse code modulation

With $L = 2^R$ levels each sample needs R bits, and at f_s samples per second the stream leaves the encoder at $R_b = Rf_s$ bits per second. **Natural binary coding** assigns 0 to $L-1$ to the levels in increasing order; **Gray coding** assigns the words so that adjacent levels differ in exactly one bit. The second is worth the trouble because a channel error almost always moves the decision to a neighbouring level, and under natural binary that can flip several bits at once — level 7 is 0111 and level 8 is 1000, four bits apart, where the Gray words 0100 and 1100 are one.

A **line code** is the rule that turns those bits into a waveform. Unipolar NRZ is on-off signalling: simplest, and it carries a DC level that makes the waveform droop through any AC-coupled stage. Polar NRZ is antipodal signalling and has no DC component for balanced data, which is why it is the shape Chapter 2 analyses. Both go quiet during a long run of identical bits, and a receiver recovering its clock from the waveform then has nothing to lock to; Manchester forces a transition in the middle of every bit, so the clock is always recoverable and there is never a DC component, paid for with twice the bandwidth.



Unipolar NRZ.



Polar NRZ, carrying the same eight bits 01101001.

EXAMPLE 1.4 — A COMPLETE PCM STREAM

GIVEN $m(t) = 8|\operatorname{sinc}(t-2)|$, so $0 \le m(t) \le 8$, sampled every $T_s = 0.6$ s and quantized by an eight-level uniform quantizer covering $[0, 8]$.

FIND The step size, the levels, the code words for $t = 0, 0.6, \ldots, 3.6$, and the bit rate.

METHOD Δ from the range and the level count; each sample by direct evaluation; each word from the tread the sample falls in.

SOLUTION $\Delta = 8/8 = 1$ V and the levels sit at $0.5, 1.5, \ldots, 7.5$. The samples are $0,\ 1.73,\ 1.87,\ 7.48,\ 6.05,\ 0,\ 1.51$, giving the words $000\ 001\ 001\ 111\ 110\ 000\ 001$. With $R = 3$ and $f_s = 1/0.6 = 1.667$, $R_b = 5$ bit/s and $T_b = T_s/3 = 0.2$ s.

CHECK The sample at $t = 3$ is $8|\operatorname{sinc}(1)| = 0$ exactly, because sinc vanishes at every non-zero integer — the same fact the interpolation formula rests on, met again at the other end of the chapter.

1.7 Summary

RESULT	STATEMENT	ANCHOR
Replication	$G_\delta(f) = f_s \sum_n G(f - nf_s)$	PS CH7.1.1
Sampling theorem	$f_s \geq 2W$ for a message bandlimited to W	PS CH7.1.1
Reconstruction	$g_r(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2Wt - n)$, filter gain $1/(2W)$	PS CH7.1.1
Step size	$\Delta = 2m_{\max}/L,\ \ L = 2^R$	PS CH7.2.1
Error power	$E[Q^2] = \Delta^2/12$, when the step is small	PS CH7.2.1
Signal-to-noise	$\mathrm{SQNR}\ [\mathrm{dB}] = \alpha + 6.02R$	PS CH7.2.1
Bit rate	$R_b = Rf_s$	PS CH7.3, 7.4.1

Sampling is reversible and quantization is not. Everything after this chapter takes the bit stream as given and asks what the channel does to it.

CHAPTER 2

Baseband transmission of digital signals

Chapter 1 produced a stream of bits. Now each bit is sent as a waveform, a channel adds noise to it, and a receiver has to say which waveform was sent. This chapter answers three questions in order: what filter the receiver should use, where the decision threshold should go, and what happens when the channel is too narrow and the pulses start to overlap.

2.1 The matched filter

Over one bit interval the receiver sees $x(t) = g(t) + w(t)$, the waveform plus white Gaussian noise of two-sided density $N_0/2$. It passes this through a filter and takes one sample at the end of the interval. Because the filter is linear, its output splits the same way, into a signal part $g_0(t) = g * h$ and a noise part $n(t) = w * h$.

The quantity worth maximising is the signal at the sampling instant against the noise power that accompanies it.

THE PEAK PULSE SIGNAL-TO-NOISE RATIO

$$(\mathrm{SNR})_o = \frac{|g_0(T)|^2}{E[n^2(t)]} = \frac{\left|\int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT}df\right|^2}{\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2 df}$$

Scaling H up multiplies the top and the bottom by the same factor, so the answer cannot be "make H large". What is being chosen is the shape of H.

Schwarz's inequality says that $\left|\int \phi_1\phi_2\right|^2 \leq \int |\phi_1|^2 \int |\phi_2|^2$, with equality only when $\phi_1 = k\phi_2^*$. Taking $\phi_1 = H(f)$ and $\phi_2 = G(f)e^{j2\pi fT}$, the factor $\int |H|^2$ cancels between the top and the bottom and a bound appears that contains no H at all.

THE BOUND, AND THE FILTER THAT REACHES IT

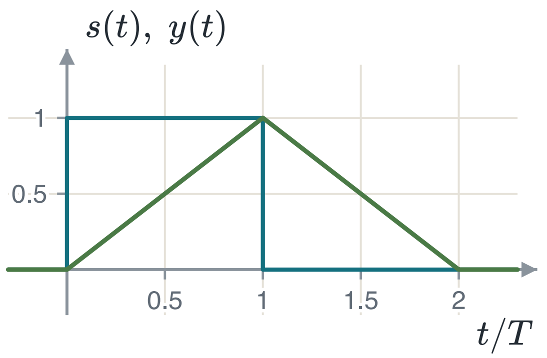
$$(\mathrm{SNR})_o \le \frac{2}{N_0}\int |G(f)|^2 df = \frac{2E}{N_0}$$

$$H(f) = k\,G^*(f)e^{-j2\pi fT} \quad \Longleftrightarrow \quad h_{\mathrm{opt}}(t) = k\,g(T-t)$$

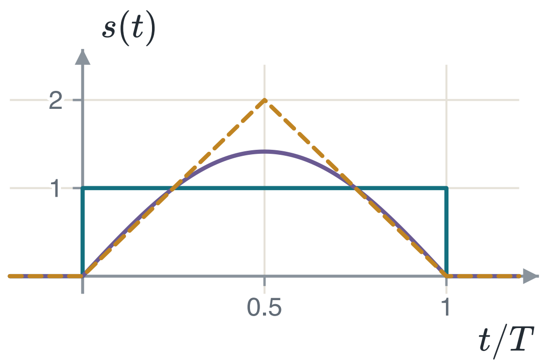
The optimum filter is the transmitted waveform reversed in time and shifted into the interval. It is called the **matched filter**.

TWO THINGS THIS SAYS

The best achievable ratio is $2E/N_0$: it depends on the **energy** of the pulse and on the noise density, and on nothing else about the pulse. Two completely different waveforms of the same energy perform identically. And the constant k cancels, so the filter is fixed only up to a gain.



A rectangular pulse and the output of the filter matched to it. The output peaks at exactly $t = T$, and its peak is the energy of the pulse.



Three pulses of the same energy. Against white Gaussian noise a matched-filter receiver performs identically on all three.

2.2 From waveform to number

Both waveforms of a binary baseband system are multiples of one shape, so a single unit-energy function carries both. With $\psi(t) = 1/\sqrt{T_b}$ on the bit interval, polar NRZ is $s_m(t) = s_m\psi(t)$ with $s_0 = -A\sqrt{T_b}$ and $s_1 = +A\sqrt{T_b}$. Both carry energy $E_b = A^2T_b$, so $A\sqrt{T_b} = \sqrt{E_b}$ and the two waveforms have become two *numbers*, $\pm\sqrt{E_b}$, on one axis.

The demodulator can be built two ways. The **matched filter** convolves with $\psi(T_b - t)$ and samples at T_b. The **correlator** multiplies by $\psi(t)$ and integrates over the interval. At the sampling instant the two produce the same number, because convolving with a reversed function and evaluating at the end of the interval *is* correlating over it.

THE DECISION STATISTIC

$$y = \int_0^{T_b} x(\tau)\psi(\tau)\,d\tau = s_m \underbrace{\int_0^{T_b}\psi^2}_{=1} + \underbrace{\int_0^{T_b} w\psi}_{n} = s_m + n$$

They agree at $t = T_b$ and nowhere else, which is one more reason the sampling instant matters.

2.3 The decision and its error probability

The noise term is a projection of a Gaussian process onto a fixed function, so it is a Gaussian random variable. Its variance follows from $E[w(\tau)w(u)] = \frac{N_0}{2}\delta(\tau - u)$ and the sifting property, and the unit energy of ψ makes the answer independent of the interval length.

WHAT THE DETECTOR IS GIVEN

$$\sigma_n^2 = \frac{N_0}{2}\int_0^{T_b} \psi^2(\tau)d\tau = \frac{N_0}{2}$$

$$y = s_m + n \sim \mathcal{N}\left(s_m, \frac{N_0}{2}\right)$$

The two-sided convention arrives here unchanged. Reading the density as one-sided makes every error probability in the course 3 dB too optimistic, and nothing in the algebra shows it.

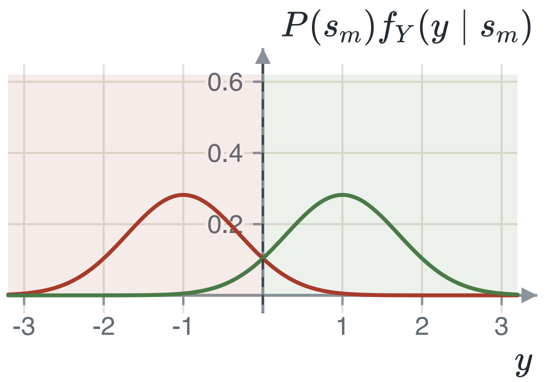
The detector decides "1" when $y > \lambda$. Writing the average error as a sum of two integrals, differentiating with respect to λ by the Leibniz rule and setting the derivative to zero gives a condition that is easy to read: the threshold sits where the two densities, each weighted by its prior, cross.

THE OPTIMAL THRESHOLD

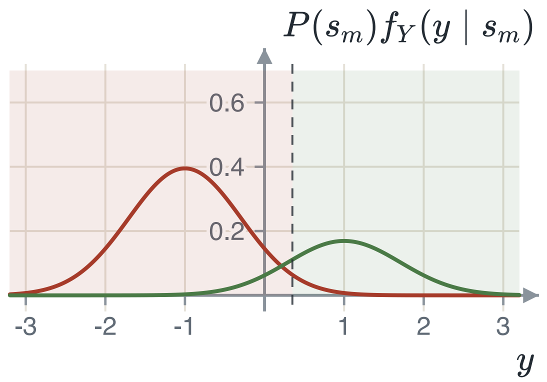
$$P(s_1)f_Y(\lambda \mid s_1) = P(s_0)f_Y(\lambda \mid s_0)$$

$$\lambda_{\text{opt}} = \frac{N_0}{4\sqrt{E_b}} \ln\frac{P(s_0)}{P(s_1)}$$

Equal priors put it midway. If s_0 is the more likely symbol the threshold moves in the positive direction, enlarging the region that decides s_0 — which is the right way round.



Equal priors: the threshold sits midway and the shaded decision regions are symmetric.



$P(s_0) = 0.7$: the more likely symbol's curve is taller and the crossing has moved towards the less likely one.

With equal priors the threshold is at the origin and both conditional errors are the same Gaussian tail, using $P(Y > y) = Q\left(\frac{y-\mu}{\sigma}\right)$ and $Q(-x) = 1 - Q(x)$.

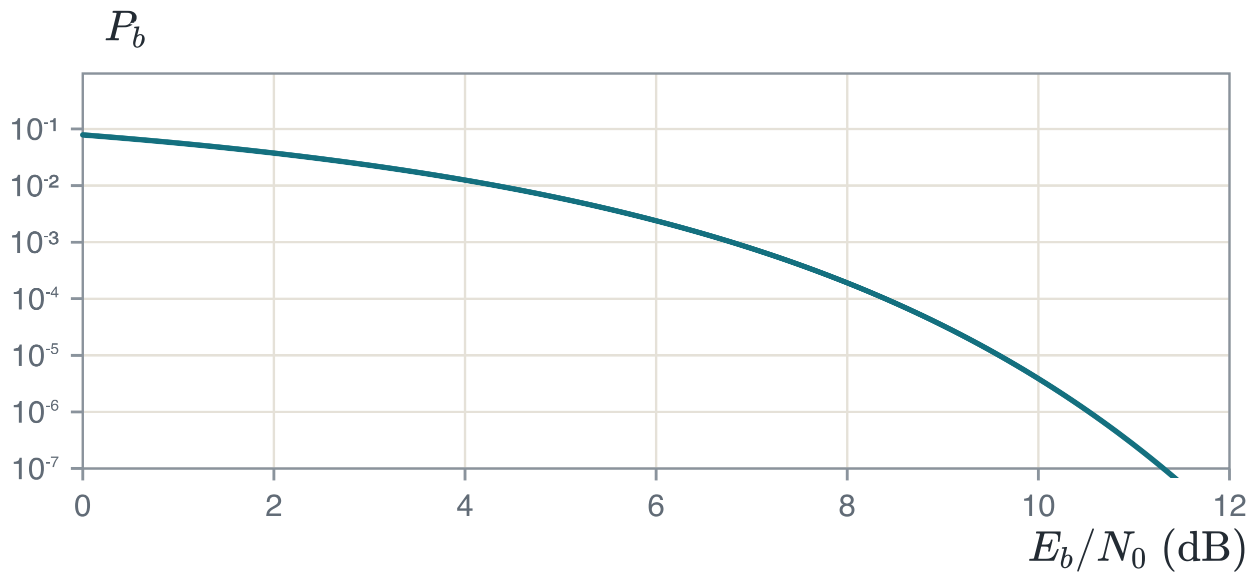
THE RESULT OF THE CHAPTER

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

It depends on E_b/N_0 and on nothing else — not on the amplitude, not on the bit duration, not on the pulse shape. Doubling the amplitude and quartering the duration change neither the energy per bit nor the answer.

THE FACTOR OF TWO THAT LIVES HERE

$Q\left(\sqrt{E_b/N_0}\right)$, the same expression without the two, is the error probability of on-off or orthogonal signalling, and it is 3 dB worse. Which one applies depends on whether the two waveforms are opposites or merely different. For on-off signalling half the bits carry no energy at all, so the average energy per bit is half the peak — and forgetting that is how on-off comes out looking as good as antipodal.



Bit error probability against E_b/N_0. Past about 8 dB every extra decibel is worth roughly an order of magnitude in error rate.

EXAMPLE 2.1 — UNEQUAL PRIORS

GIVEN A binary PAM system with correlator output $y = \pm\sqrt{E_b} + n$, where $P(s_1) = 0.3$, $E_b = 1$ and $N_0 = 0.1$.

FIND The optimal threshold and the average error probability there.

METHOD Threshold from the log-ratio of the priors; each conditional error from a Gaussian tail; average by weighting.

SOLUTION $\lambda = \frac{0.1}{4} \ln \frac{0.7}{0.3} = 0.0212$ and $\sigma = \sqrt{0.05} = 0.2236$. Then $P(\mathrm{err} \mid s_0) = Q(4.567) = 2.475 \times 10^{-6}$ and $P(\mathrm{err} \mid s_1) = Q(4.377) = 6.005 \times 10^{-6}$, so $P_e = 0.7(2.475) + 0.3(6.005)$ in units of 10^{-6}, giving 3.534×10^{-6}.

CHECK Leaving the threshold at zero would give $Q(\sqrt{20}) = 3.872 \times 10^{-6}$. Moving it improves the answer by about nine per cent — a real gain and a small one, which is what a shift of a tenth of a standard deviation buys.

2.4 Intersymbol interference

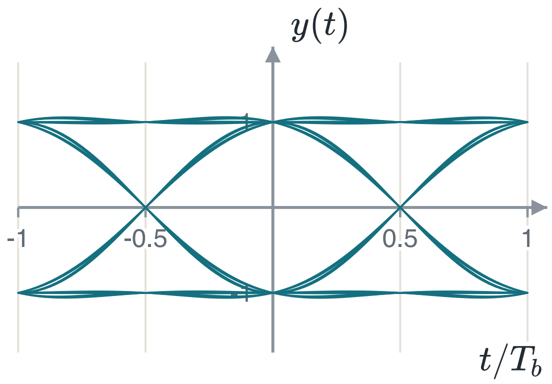
Everything above assumed an ideal channel. A real channel is bandlimited, and a bandlimited channel spreads each pulse in time, so a pulse meant for one interval leaks into the next. Writing the received signal as a train of the overall pulse p and sampling at $t_i = iT_b$ separates the term that was meant to be there from the ones that were not.

WHERE THE INTERFERENCE COMES FROM

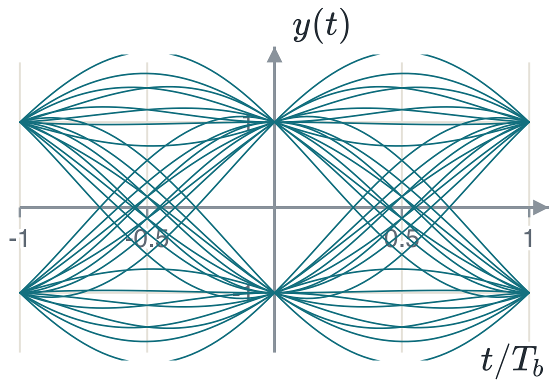
$$y(t_i) = \underbrace{\mu a_i}_{\text{wanted}} + \underbrace{\mu \sum_{k \neq i} a_k \, p\big((i-k)T_b\big)}_{\text{intersymbol interference}} + n(t_i)$$

The middle term is not noise. It is caused by the data itself, and raising the transmit power raises it by the same factor, so more power does not help. At high signal-to-noise ratio it is the only thing limiting the system.

The **eye pattern** is how this is seen rather than calculated. Cut the received waveform into bit-length pieces, draw them all on top of each other synchronised to the clock, and four measurements are readable at once: the height of the opening is the margin over noise, its width is how far the sampling instant may drift, the slope at the crossings is the sensitivity to a timing error, and the spread of the crossings is the timing jitter. When the opening closes there is no sampling instant at which every bit pattern decides correctly, and no threshold placement helps.



A wide-open eye: a pulse that decays quickly. Every trace passes close to ± 1 at the centre.



A slowly decaying pulse. The opening has narrowed and the crossings have scattered, with the noise unchanged.

2.5 Nyquist's criterion and the raised cosine

The interference vanishes exactly when the overall pulse is zero at every sampling instant but its own. Sampling p at those instants and transforming turns that requirement into a statement about the spectrum.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

$$p\big((i-k)T_b\big) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

$$R_b \sum_n P(f - nR_b) = 1, \qquad R_b = \frac{1}{T_b}$$

In words: the replicas of the pulse spectrum, spaced by the symbol rate, must add to a constant. The simplest spectrum that does it is a rectangle of width $2W$ with $W = R_b/2$, the **Nyquist bandwidth**, whose pulse is $\mathrm{sinc}(2Wt)$.

THE RATE A BANDWIDTH SUPPORTS

A channel of bandwidth W carries at most $2W$ symbols per second with no interference. This is the counterpart of the sampling theorem, reached from the other end of the same argument.

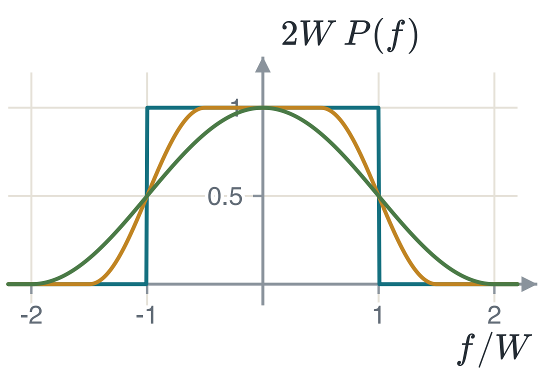
The ideal Nyquist channel cannot be built: its spectrum has vertical edges, and its pulse decays only as $1/t$, so a small timing error lets a long tail of neighbours contribute at once. The fix is to widen the spectrum and taper its edges while keeping the tiling property.

THE RAISED COSINE

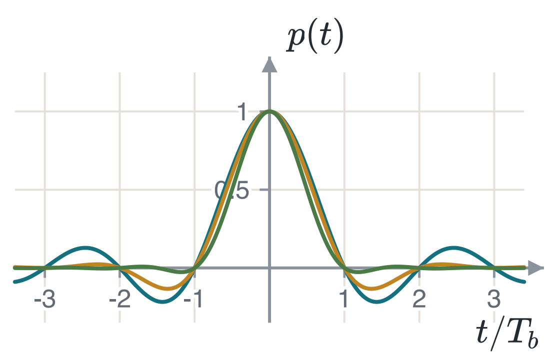
$$P(f) = \begin{cases} \dfrac{1}{2W}, & 0 \le |f| \le f_1 \\ \dfrac{1}{4W}\left[1 - \sin\dfrac{\pi(|f| - W)}{2W - 2f_1}\right], & f_1 \le |f| < 2W - f_1 \\ 0, & \text{otherwise} \end{cases}$$

$$p(t) = \mathrm{sinc}(2Wt)\,\frac{\cos(2\pi\alpha Wt)}{1 - 16\alpha^2W^2t^2}, \qquad \alpha = 1 - \frac{f_1}{W}$$

The first factor keeps the zero crossings at $t = iT_b$, so the criterion still holds; the second decays as $1/t^2$. The price is bandwidth: $B_T = (1+\alpha)W$, an excess of αW over Nyquist.



The spectrum at $\alpha = 0$, 0.5 and 1. All three tile the axis at spacing $2W$, so all three give zero interference.



Their pulses. All three vanish at every non-zero multiple of T_b; the tails differ by orders of magnitude.

EXAMPLE 2.2 — FITTING A RATE INTO A CHANNEL

GIVEN	A channel of bandwidth 48 kHz is to carry 64 kbit/s with raised-cosine shaping.
FIND	The Nyquist bandwidth and the largest roll-off that fits.
METHOD	$W = R_b/2$, then $(1+\alpha)W \le B$.
SOLUTION	$W = 32$ kHz, and $(1+\alpha)(32) \le 48$ gives $\alpha \le 0.5$. At $\alpha = 0.5$ the signal uses the whole 48 kHz, with an excess of 16 kHz over the Nyquist bandwidth.
CHECK	At $\alpha = 0$ the signal would need only 32 kHz and would also fit — and would be a bad choice, because its pulse decays as $1/t$ and its filter cannot be built. Using the whole channel buys a pulse decaying as $1/t^3$ and a filter that exists. The spectral efficiency is $64/48 = 1.33$ bit/s/Hz against the theoretical 2.

2.6 Summary

RESULT	STATEMENT	ANCHOR
Matched filter	$h_{\text{opt}}(t) = g(T-t)$	PS CH8.3.2
What it achieves	$(\text{SNR})_o = 2E/N_0$, independent of the pulse shape	PS CH8.3.2
Decision statistic	$y = s_m + n,\ \ n \sim \mathcal{N}(0, N_0/2)$	PS CH8.3.1
Optimal threshold	$\lambda = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$	PS CH8.3.3
Antipodal error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	PS CH8.3.3
On-off error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$, three decibels worse	PS CH8.3.3
Nyquist criterion	$R_b \sum_n P(f - nR_b) = 1$	PS CH10.3.1
Raised cosine	$B_T = (1+\alpha)W$	PS CH10.3.1

Two waveforms have become two points on a line, and the error probability has turned out to depend on the distance between them. Chapter 3 asks what happens when there are more than two waveforms, and finds that the same picture works with more axes.

CHAPTER 3

Geometric representation of signal waveforms

This chapter changes the language, not the subject. A set of waveforms is written as a set of points. Once that is done, the receiver becomes a question about geometry, and the answers are arithmetic instead of integrals.

3.1 Why one axis is not enough

In Chapter 2 the two waveforms were opposites. One shape carried both, the receiver produced one number, and everything followed from that. Now suppose the two waveforms are simply different: neither is a multiple of the other. No single function can write both as a multiple of itself, so the receiver of Chapter 2 does not apply.

The fix is to use two matched filters, one for each waveform, and let the receiver work with the two numbers they produce. A pair of numbers is a point in a plane. That plane is the subject of this chapter.

3.2 Signals as vectors

A vector is a list of numbers because it has been written against a set of axes. A signal can be a list of numbers for the same reason. The only thing needed is an inner product, and for signals it is the integral of the product.

INNER PRODUCT, ENERGY AND ORTHOGONALITY

$$\langle x, y\rangle = \int_{-\infty}^{\infty} x(t)\, y(t)\, dt$$

$$\int_{-\infty}^{\infty} \psi_j(t)\psi_k(t)\, dt = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

The second line defines an **orthonormal** set. The first case says each function has unit energy; the second says any two of them are orthogonal.

WHAT ORTHOGONAL MEANS HERE

The same thing it means for arrows: the two signals have nothing of each other in them. Multiply them together, add up the result over all time, and you get zero. Two pulses that never overlap are the clearest case — where one is non-zero the other is zero, so the product is zero everywhere and no integration is needed.

With the axes fixed, a signal is written against them the same way a vector is. One integral per axis takes the waveform apart; adding the pieces back builds it again.

COORDINATES

$$s_{ij} = \int_0^T s_i(t)\, \psi_j(t)\, dt$$

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\, \psi_j(t)$$

The list $\mathbf{s}_i = (s_{i1}, \ldots, s_{iN})$ is the **signal vector**.

The reason this is worth doing is that inner products survive the translation. Expand both signals in the basis, exchange the integral with the sums, and use orthonormality: every cross term vanishes and every matching term contributes one. An integral over waveforms becomes a sum over N numbers.

THE PROPERTY THE CHAPTER RESTS ON

$$\int_{-\infty}^{\infty} x(t)y(t)\,dt = \sum_{k=1}^{N} x_k y_k$$

$$E_{s_i} = \|\mathbf{s}_i\|^2 = \sum_{j=1}^{N} s_{ij}^2, \qquad \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \int_0^T \big(s_i - s_k\big)^2 dt$$

Energy is how far a point is from the origin. That is what the transmitter pays for. **Distance is how far two points are from each other.** That is what decides how often the receiver confuses them.

3.3 The constellation diagram

The constellation is the picture of the signal vectors in the space their basis spans: one point per waveform, one axis per basis function. Three things are read straight off it. The distance from the origin to a point is the square root of that signal's energy. The distance between two points is the square root of the energy of their difference. And the number of axes is the number of matched filters the receiver needs.

TWO STATEMENTS, AND WHAT THEY MEAN TOGETHER

Two entirely different signal sets can have the same geometric representation. Different shapes, different durations, the same points. And **the geometry determines the performance and the receiver structure**. Put together: if two signal sets give the same constellation, they have the same error probability and the same receiver, and nothing about the waveforms themselves survives into the answer. What the picture does *not* fix is bandwidth, and that is decided by the criteria of Chapter 2.

3.4 The Gram–Schmidt procedure

The axes have been assumed so far. Gram–Schmidt produces them from any set of waveforms. It is one step repeated: take the next signal, remove the part of it that the axes already found can account for, and make an axis out of what is left.

THE PROCEDURE

$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}}, \qquad s_{11} = \sqrt{E_1}$$

$$g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki}\psi_i(t), \qquad \psi_k(t) = \frac{g_k(t)}{\sqrt{E_{g_k}}}$$

If $g_k(t) = 0$ the signal was already a combination of the axes found so far and **no new axis is added**. The number of axes N is therefore at most the number of signals M, and is often fewer.

THE ORDER CHANGES THE AXES, NOT THE ANSWER

Starting from a different signal gives a different basis. It gives the same number of axes, the same energies and the same distances, so the constellation is the same picture seen from a different angle. Every result that depends only on distances is unchanged — and in Chapter 4 every result depends only on distances.

EXAMPLE 3.1 — THREE PULSES, TWO AXES

GIVEN $s_1(t) = 1$ on $[0, 2]$; $s_2(t) = 1$ on $[2, 3]$; $s_3(t) = 1$ on $[0, 3]$; each zero elsewhere.

FIND An orthonormal basis, the three signal vectors, and the constellation.

METHOD Normalise the first signal. Then, for each later signal, subtract the part that lies along the axes already found and normalise the remainder.

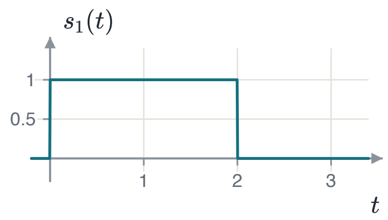
STEP 1 $E_1 = \int_0^2 1\,dt = 2$, so $\psi_1(t) = 1/\sqrt{2}$ on $[0, 2]$ and $s_{11} = \sqrt{2}$.

STEP 2 $s_{21} = \int s_2\psi_1\,dt = 0$, because the two never overlap. So $g_2 = s_2$, its energy is 1, and $\psi_2(t) = s_2(t)$.

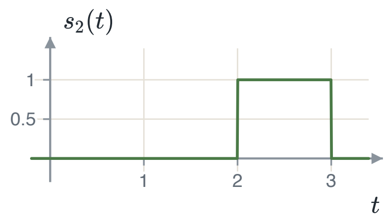
STEP 3 $s_{31} = \sqrt{2}$ and $s_{32} = 1$, and then $g_3 = s_3 - \sqrt{2}\psi_1 - \psi_2 = 0$. No third axis is added.

VECTORS $\mathbf{s}_1 = (\sqrt{2}, 0),\ \ \mathbf{s}_2 = (0, 1),\ \ \mathbf{s}_3 = (\sqrt{2}, 1)$.

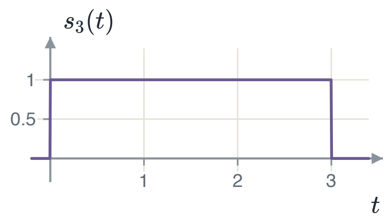
CHECK Energies from the vectors are 2, 1 and 3. From the waveforms: height one for two seconds, one second and three seconds, giving 2, 1 and 3. They agree, and that agreement is the whole content of "energy is squared length". Three waveforms needed only two axes, because $s_3 = s_1 + s_2$ — visible in the pictures before any integral is computed.



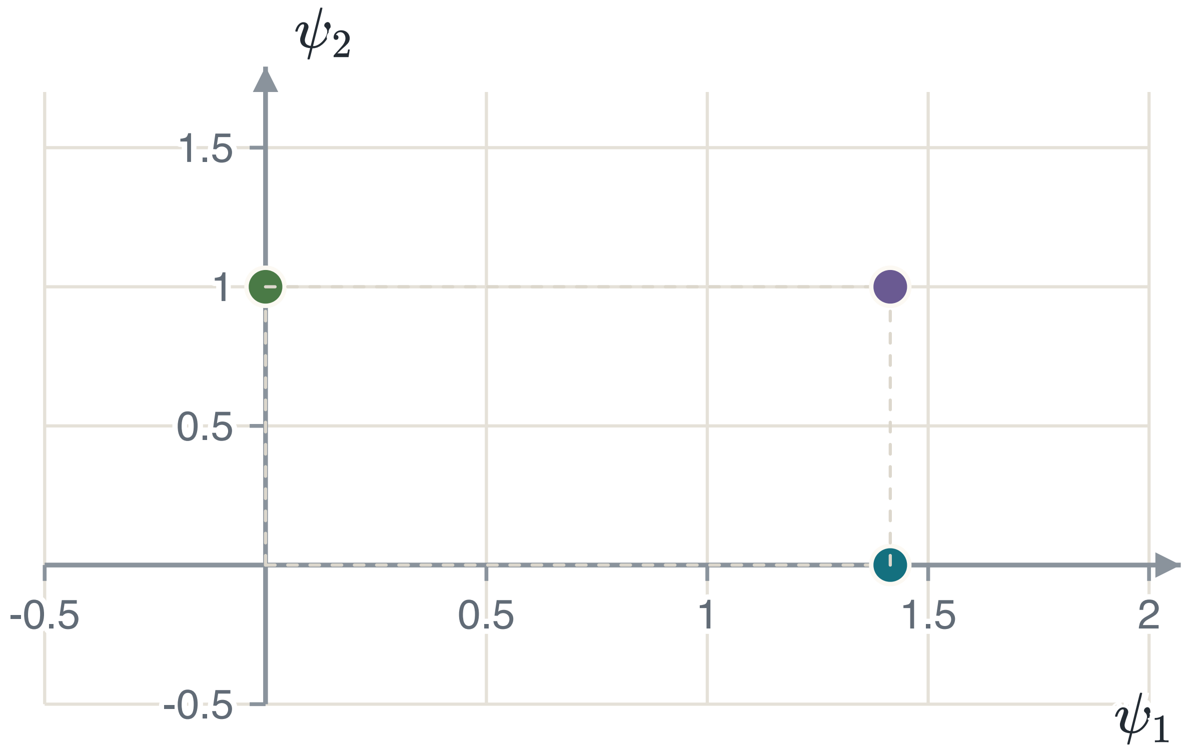
s_1



s_2



$s_3 = s_1 + s_2$



The constellation. Two axes carry three signals. The point furthest from the origin is $\mathbf{s}_3$, whose energy is 3; the two nearest points are $\mathbf{s}_1$ and $\mathbf{s}_3$, one unit apart.

3.5 Summary

RESULT	STATEMENT	ANCHOR
Orthonormal set	$\int \psi_j \psi_k \, dt = 1$ if $j = k$, else 0	PS CH8.1
Coordinates	$s_{ij} = \int s_i \psi_j \, dt$	PS CH8.1
Inner product survives	$\int xy \, dt = \sum_k x_k y_k$	PS CH8.1
Energy	$E_i = \|\mathbf{s}_i\|^2$	PS CH8.1
Distance	$\|\mathbf{s}_i - \mathbf{s}_k\|^2 =$ energy of the difference	PS CH8.1
Gram–Schmidt	$g_k = s_k - \sum_{i<k} s_{ki} \psi_i$, and $\psi_k = g_k / \sqrt{E_{g_k}}$	PS CH8.1

Chapter 4 builds the receiver for this picture. It computes the coordinates of whatever arrives and picks the nearest point, and the error probability then depends on the distances between the points and on nothing else.

CHAPTER 4

The optimal receiver in additive white Gaussian noise

A transmitter picks one of M signals and sends it; the channel adds noise; the receiver has to name the signal. Chapter 3 turned the signals into points. This chapter turns the receiver into a rule about those points, and the rule is simple: choose the point nearest to what arrived.

4.1 What the receiver keeps

The received signal is $r(t) = s_i(t) + n(t)$ over one symbol interval. The receiver has N correlators, one for each basis function, and each returns one number.

THE OBSERVATION VECTOR

$$r_k = \int_0^T r(t)\psi_k(t)\,dt = s_{ik} + n_k$$

$$\mathbf{r} = \mathbf{s}_i + \mathbf{n} = (r_1, \ldots, r_N)$$

The signal was built from the N basis functions, so the correlators capture all of it. The noise was not: the part of it outside the signal space, written $n_0(t)$, is thrown away.

WHY THROWING $n_0(t)$ AWAY COSTS NOTHING

It contains no signal — every signal is a combination of the N basis functions and has no component outside that space. And it is independent of the N numbers the receiver kept. Something that carries no information about the answer and is unrelated to what was kept cannot help. Note what this does *not* say: the noise inside the signal space is kept in full, and it is all the trouble there is.

Each n_k is a projection of a Gaussian process onto a fixed function, so it is Gaussian with zero mean. The useful part is what happens between two of them: using $E[n(\tau)n(u)] = \frac{N_0}{2}\delta(\tau - u)$ and orthonormality, $E[n_j n_k]$ is $N_0/2$ when $j = k$ and zero otherwise. Uncorrelated jointly Gaussian variables are independent, so the noise components are independent, each $\mathcal{N}(0, N_0/2)$.

WHAT THAT MEANS FOR THE PICTURE

Every axis carries the same variance and the axes are independent, so the noise cloud is a **circle** — a sphere in N dimensions — with no preferred direction. That symmetry is why the receiver may simply measure distance. If the variances differed, the nearest point in the ordinary sense would not be the best answer.

4.2 The decision rule

The receiver should choose the signal that is most likely given what it saw. By Bayes' rule that means maximising $P(\mathbf{s}_i) f_{\mathbf{r}}(\mathbf{r} \mid \mathbf{s}_i)$ — the prior times the likelihood. This is the **MAP** rule, and no rule has a smaller error probability. If all M signals are equally likely the priors cannot change the answer, and what is left is the **ML** rule: maximise the likelihood alone.

Putting the noise density into the likelihood and taking a logarithm — which is increasing, so it changes no winner — turns the product into a sum and leaves one term that depends on i.

MINIMUM-DISTANCE DETECTION

$$\ln f_{\mathbf{r}}(\mathbf{r} \mid \mathbf{s}_i) = -\frac{N}{2}\ln(\pi N_0) - \frac{1}{N_0}\|\mathbf{r} - \mathbf{s}_i\|^2$$

$$\hat{s} = \arg\min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad \text{(ML)}$$

$$\hat{s} = \arg\min_i \Big\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \Big\} \quad \text{(MAP)}$$

In words: choose the signal point closest to what arrived. With unequal priors, subtract a fixed handicap from each distance first — the term is positive and larger for the more likely symbol, so that symbol wins from further away.

Expanding the squared distance gives the form a receiver is actually built in. The term $\|\mathbf{r}\|^2$ is the same for every i and drops out; turning the minimum into a maximum and dividing by two leaves the **correlation metric**.

THE RECEIVER AS IT IS BUILT

$$\hat{s} = \arg\max_i \Big\{ \mathbf{r}\cdot\mathbf{s}_i - \frac{E_i}{2} + \frac{N_0}{2}\ln P(\mathbf{s}_i) \Big\}$$

And $\mathbf{r} \cdot \mathbf{s}_i = \int_0^T r(t)s_i(t)\,dt$, so the receiver can correlate against the waveforms directly and never compute coordinates. If the signals are equally likely **and** have equal energy, both corrections vanish and the rule is: take the largest correlation.

WHEN THE ENERGIES DIFFER, KEEP THE ENERGY TERM

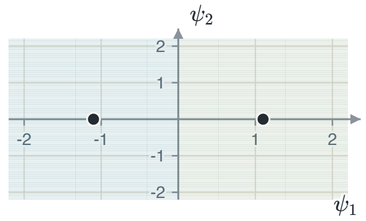
Without $-E_i/2$ a high-energy signal wins too often, because a large $\mathbf{s}_i$ makes $\mathbf{r} \cdot \mathbf{s}_i$ large whatever arrived. On-off signalling is exactly this case, and dropping the term there makes the receiver decide "one" almost always. Equal priors remove the prior term; only equal *energies* remove the energy term.

4.3 Decision regions

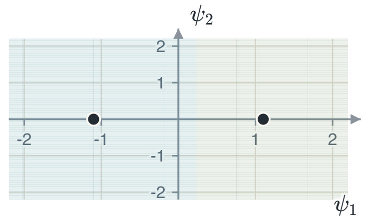
Collecting the observations that give the same answer divides the space into M regions, $R_i = \{\mathbf{r} : \|\mathbf{r} - \mathbf{s}_i\| \le \|\mathbf{r} - \mathbf{s}_j\| \text{ for all } j\}$. Three facts describe them, and all three follow from the rule being "nearest point".

1. A boundary between two points is **perpendicular** to the line joining them.
2. With equal priors it crosses that line exactly **halfway**.
3. With unequal priors it moves, and **the region of the less likely signal shrinks**.

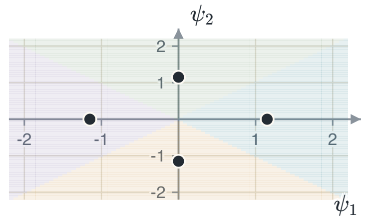
The first two are the definition of a perpendicular bisector: the set of points equidistant from two fixed points *is* that bisector. Nothing has to be calculated. Only nearest neighbours contribute boundaries, so a point in the middle of a constellation has a bounded region and a point on the outside has one that runs off to infinity — which is why outer points make fewer errors.



Two points, equal priors: the boundary is halfway.



The left symbol four times more likely: its region has grown.



Four points, four boundaries, each perpendicular to the pair it separates.

For two equally likely points a distance d apart, the boundary is $d/2$ from each. An error happens when the noise along the line joining them carries the observation across, and the noise on any one axis is $\mathcal{N}(0, N_0/2)$.

THE BINARY RESULT, FROM THE PICTURE

$$P_e = Q\left(\frac{d/2}{\sqrt{N_0/2}}\right) = Q\left(\sqrt{\frac{d^2}{2N_0}}\right)$$

$$\mu = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(\mathbf{s}_1)}{P(\mathbf{s}_0)}$$

The second line is where the boundary sits when the priors are unequal, measured from the first point along the line. Equal priors give $\mu = d/2$.

THE RESULT THIS CHAPTER EXISTS FOR

The error probability of a binary system depends on **the distance between the two points and on nothing else**. Antipodal points are $2\sqrt{E_b}$ apart, giving $Q\left(\sqrt{2E_b/N_0}\right)$; on-off or orthogonal points are $\sqrt{2E_b}$ apart, smaller by $\sqrt{2}$ — which is the 3 dB of Chapter 2, obtained here by measuring a picture instead of integrating two densities.

4.4 The union bound

For more than two signals the exact answer is an integral of the likelihood over each decision region, in N dimensions and over a polygon with as many faces as the point has neighbours. There is no closed form. The way round it is to bound the answer instead.

Suppose $\mathbf{s}_k$ was sent, and let A_{kj} be the event that the observation is closer to $\mathbf{s}_j$ than to $\mathbf{s}_k$. An error happens exactly when at least one of those occurs, and the probability of a union is at most the sum of the probabilities. Each term is a two-point question whose answer is already known.

THE UNION BOUND AND ITS TWO USABLE FORMS

$$P(\mathbf{s}_k \to \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le \frac{1}{M}\sum_k \sum_{j\ne k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le (M-1)Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \qquad P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

$N_{\min}$ is the *average* number of points at the minimum distance, and need not be an integer. The last form is the one used in practice.

WHY IT IS AN OVER-ESTIMATE, AND WHEN THAT MATTERS

The events overlap: an observation can be closer to two other points at once, and the sum counts it twice. So the bound is always at least the truth. At low signal-to-noise ratio the overlaps are large and the bound can exceed one, which is useless; at the ratios real systems run at, every term is tiny and the bound is close enough to be quoted as the answer.

EXAMPLE 4.1 — THREE BOUNDS ON ONE CONSTELLATION

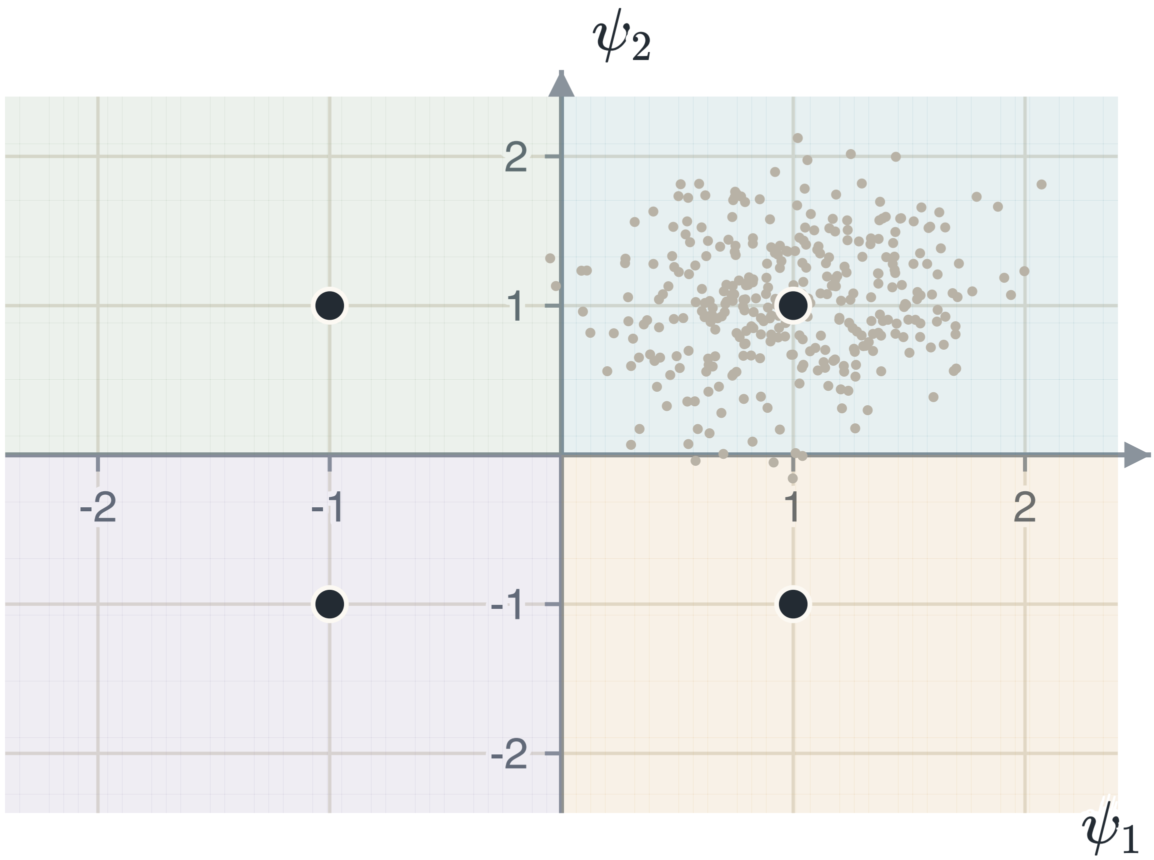
GIVEN: Four equally likely points at $(\pm d/2, \pm d/2)$: neighbours d apart, diagonals $d\sqrt{2}$.

FIND: The symbol error probability by each of the three forms.

METHOD: Take one point, list its distances to the other three, and add one Q for each. Symmetry makes every point give the same answer.

SOLUTION: General: $P_e \le 2Q\left(\sqrt{d^2/2N_0}\right) + Q\left(\sqrt{2d^2/2N_0}\right)$. Nearest neighbours ($d_{\min} = d$, $N_{\min} = 2$): $P_e \approx 2Q\left(\sqrt{d^2/2N_0}\right)$. Minimum distance ($M-1=3$): $P_e \le 3Q\left(\sqrt{d^2/2N_0}\right)$.

CHECK: Put $d^2/2N_0 = 9$, so $Q(3) = 1.35 \times 10^{-3}$ and $Q(4.243) = 1.10 \times 10^{-5}$. The three answers are 2.71×10^{-3}, 2.70×10^{-3} and 4.05×10^{-3}: the diagonal term is worth 0.4%, and pretending it is a nearest neighbour costs 50%. All three are the same Q with a different count in front, so none of them moves the curve sideways.



The constellation, its four regions, and the observations the receiver sees when the top-right point is sent. The exact error probability is the fraction of that cloud outside its own region; the bound adds three separate two-point questions and counts the overlaps twice.

4.5 Summary

RESULT	STATEMENT	ANCHOR
Observation	$\mathbf{r} = \mathbf{s}_i + \mathbf{n}$, each $n_k \sim \mathcal{N}(0, N_0/2)$, independent	PS CH8.4.1
MAP rule	minimise $\lVert\mathbf{r} - \mathbf{s}_i\rVert^2 - N_0 \ln P(\mathbf{s}_i)$	PS CH8.4.1
ML rule	minimise $\lVert\mathbf{r} - \mathbf{s}_i\rVert^2$: choose the nearest point	PS CH8.4.1
Receiver	maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2 + \frac{N_0}{2} \ln P(\mathbf{s}_i)$	PS CH8.4.1
Regions	perpendicular bisectors; the less likely region shrinks	PS CH8.4.1
Binary	$P_e = Q\left(\sqrt{d^2/2N_0}\right)$	PS CH8.3.3
Union bound	$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q(\cdot)$	PS CH8.4.2
In practice	$P_e \approx N_{\min} Q\left(\sqrt{d_{\min}^2/2N_0}\right)$	PS CH8.4.2

A constellation is judged by two numbers: $d_{\min}$, which sets the exponent and therefore almost everything, and $N_{\min}$, which multiplies it and matters far less. Chapter 5 applies this to the constellations that are actually used, and asks which of them gets the largest $d_{\min}$ for a given average energy and a given number of bits per symbol.

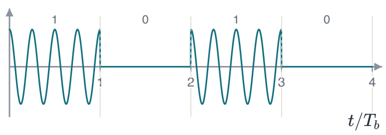
CHAPTER 5

Digital modulation methods

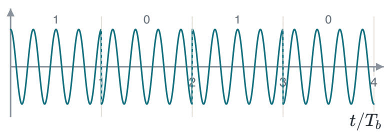
A baseband pulse cannot be sent over a radio channel, so the bits have to ride on a carrier. There are only three things about a sine wave that can be changed: its amplitude, its phase and its frequency. Every scheme in this chapter changes one of them, or two at once. Chapter 4 already gave the receiver and the error probability; all that is left is to place the points and measure the distance between them.

5.1 The three binary schemes

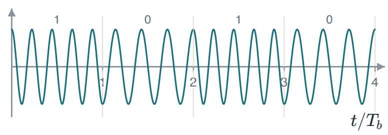
Take one bit at a time and two waveforms. Amplitude-shift keying sends the carrier for a one and nothing for a zero. Phase-shift keying sends the carrier either way and flips its sign. Frequency-shift keying sends one of two frequencies.



Amplitude-shift keying: the carrier is switched on and off.



Phase-shift keying: the carrier is always on and its sign flips.



Frequency-shift keying: one frequency for a one, another for a zero.

To compare them, put each on the axes of chapter 3 and measure. BPSK needs one basis function and its two points are $\pm\sqrt{E_b}$ on it. BFSK needs two, because the two frequencies are orthogonal, and its points sit one on each axis. BASK needs one, and its points are 0 and $\sqrt{2E_b}$ once the average over equally likely bits is taken.

THE THREE DISTANCES AND THE THREE ERROR PROBABILITIES

$$\text{BPSK: } d_{\min}^2 = 4E_b, \qquad P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{BFSK and BASK: } d_{\min}^2 = 2E_b, \qquad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

The two answers differ by a factor of two inside the square root, which is 3.01 dB. That gap does not depend on the noise level and cannot be closed by spending more power — it is a fact about where the points are.

WHERE THE THREE DECIBELS COME FROM

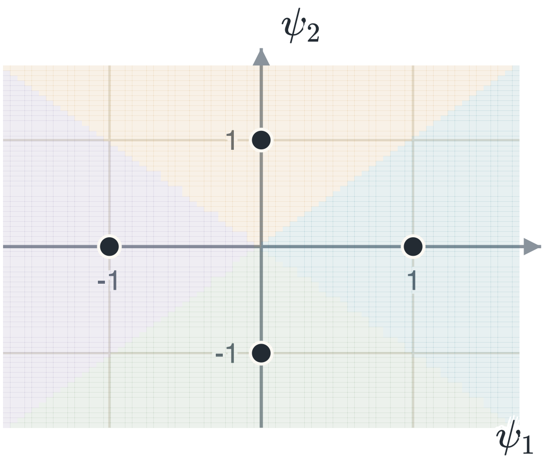
Antipodal points are $2\sqrt{E_b}$ apart. Orthogonal points are $\sqrt{2E_b}$ apart, which is smaller by $\sqrt{2}$. Squaring that $\sqrt{2}$ gives the factor of two in the Q argument, and $10\log_{10} 2 = 3.01$ dB. Every appearance of three decibels in this chapter traces back to one $\sqrt{2}$ in a picture.

BASK AND BFSK REACH THE SAME ANSWER BY DIFFERENT ROUTES

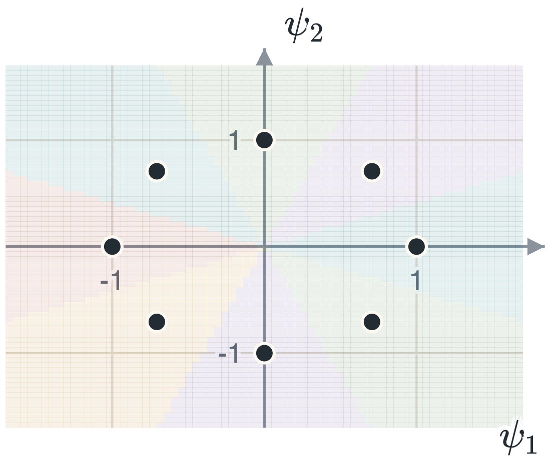
BASK keeps its points on one line but half its symbols cost nothing, so the average energy is half the peak. BFSK keeps the full energy in every symbol but separates the points by a right angle instead of a straight line. Both end at $d^2 = 2E_b$. The mathematics does not prefer either; the transmitter does, and BASK asks it to switch between zero power and full power.

5.2 M-ary phase-shift keying

Sending one bit at a time wastes the two dimensions a carrier offers. Use both, keep the energy fixed, and the constellation becomes M points spaced evenly around a circle of radius $\sqrt{E_s}$. Each symbol now carries $\log_2 M$ bits.



$M = 4$, two bits a symbol. The four regions are the four quadrants.



$M = 8$, three bits. The regions are wedges, and they are narrowing.

Two neighbouring points subtend an angle $2\pi/M$ at the centre. Dropping a perpendicular from the centre onto the line joining them cuts that angle in half and gives the distance directly.

M–PSK

$$d_{\min} = 2\sqrt{E_s}\,\sin\left(\frac{\pi}{M}\right), \qquad N_{\min} = 2$$

$$P_e \approx 2\,Q\left(\sqrt{\frac{2E_s}{N_0}}\,\sin\frac{\pi}{M}\right)$$

$N_{\min} = 2$ for every M above two, because a point on a circle has one neighbour clockwise and one anticlockwise. At $M = 2$ the formula gives $d_{\min} = 2\sqrt{E_s}$, which is BPSK — a general result that reproduces the case already known.

EXAMPLE 5.1 — WHAT THE STEP FROM FOUR POINTS TO EIGHT COSTS

GIVEN QPSK and 8-PSK, at whatever energy each needs to reach the same symbol error probability.

FIND The extra energy 8-PSK needs, per symbol and per bit.

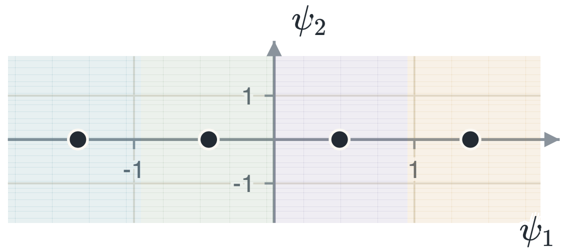
METHOD A fixed error probability means a fixed Q argument, so $(E_s/N_0)\sin^2(\pi/M)$ must be held constant. Take the ratio.

SOLUTION $\dfrac{\sin^2(\pi/4)}{\sin^2(\pi/8)} = \dfrac{0.5}{0.1464} = 3.414$, which is 5.33 dB per symbol. 8-PSK carries three bits where QPSK carries two, so per bit the figure is $3.414 \times \frac{2}{3} = 2.276$, or 3.57 dB.

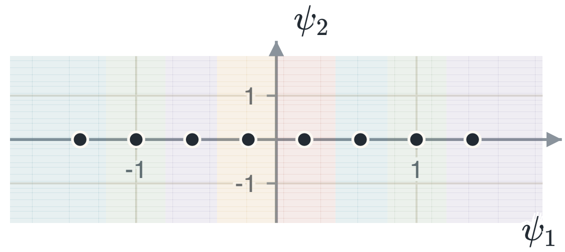
CHECK Both numbers are positive, as they must be: at the same radius, eight points are closer together than four. The per-bit figure is the smaller of the two, because the extra bit pays part of its own way.

5.3 M-ary amplitude-shift keying

Change the amplitude among M levels instead. One basis function carries them all, so the constellation is M points on a line at $\pm A, \pm 3A, \dots, \pm(M-1)A$, and neighbouring points are $2A$ apart.



$M = 4$ on one axis. The two end regions run off to infinity.



$M = 8$: the same line, twice as crowded.

M-PAM

$$E_{s,\text{avg}} = \frac{A^2(M^2-1)}{3} \quad\Longrightarrow\quad d_{\min}^2 = \frac{12E_{s,\text{avg}}}{M^2-1}$$

$$N_{\min} = \frac{2(M-1)}{M}, \qquad P_e \approx N_{\min}\,Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

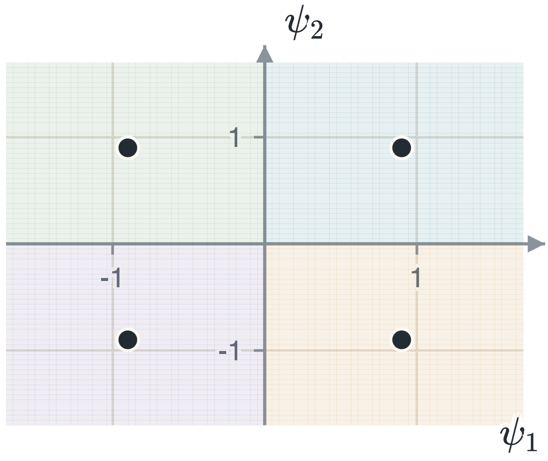
The M^2 in the denominator is the trouble: doubling M divides the squared distance by roughly four, which is about 6 dB, and it does so at every doubling.

WHY $N_{\min}$ IS NOT A WHOLE NUMBER

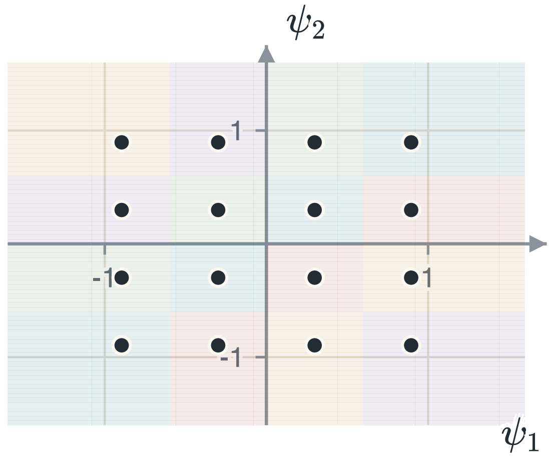
The two points at the ends of the line have one neighbour each; the $M-2$ points between them have two. Averaging over all M equally likely symbols gives $2(M-1)/M - 1.5$ at $M = 4$, 1.75 at $M = 8$. It is an average over points that differ in how exposed they are, and getting a whole number means the end points were counted as though they had neighbours on both sides.

5.4 Quadrature amplitude modulation

Run one amplitude-modulated signal on the cosine and an independent one on the sine. The result is two $\sqrt{M}$-level constellations at right angles, so M points on a square grid — the lecture material states this as M-QAM being two-dimensional M-ASK.



4-QAM: two levels a side. These are the four points of QPSK — at $M = 4$ the two schemes are the same scheme.



16-QAM: four levels a side, four bits a symbol. The corners have two neighbours, the edges three, the four in the middle four.

SQUARE M-QAM

$$E_{s,\text{avg}} = \frac{(M-1)d^2}{6} \implies d^2_{\text{min}} = \frac{6E_{s,\text{avg}}}{M-1}$$

$$N_{\text{min}} = \frac{4(2)+8(3)+4(4)}{16} = 3 \quad \text{for 16-QAM}$$

Compare the denominators: $M-1$ for QAM against M^2-1 for PAM. Spreading the same number of points over two dimensions instead of one turns a square into a linear factor, and that is the whole of the difference.

EXAMPLE 5.2 — SIXTEEN POINTS ON A GRID AGAINST SIXTEEN ON A LINE

GIVEN 16-QAM and 16-PAM at the same average symbol energy. Both carry four bits a symbol.

FIND The advantage of QAM in decibels.

METHOD One formula each, then the ratio of the squared distances.

SOLUTION QAM: $d^2_{\text{min}} = 6E_s/15 = 0.400E_s$. PAM: $d^2_{\text{min}} = 12E_s/255 = 0.0471E_s$. The ratio is 8.5, and $10\log_{10} 8.5 = 9.29$ dB.

CHECK Same number of bits, same average power, and the only difference is where the points were put. Nine decibels is an enormous return for using a second dimension that was there all along, and it is the reason QAM is what modern systems use.

5.5 M-ary frequency-shift keying

Use M frequencies, chosen so that the waveforms are orthogonal. Orthogonal signals need one basis function each, so the constellation lives in M dimensions with one point on each axis at distance $\sqrt{E_s}$ from the origin.

M–FSK

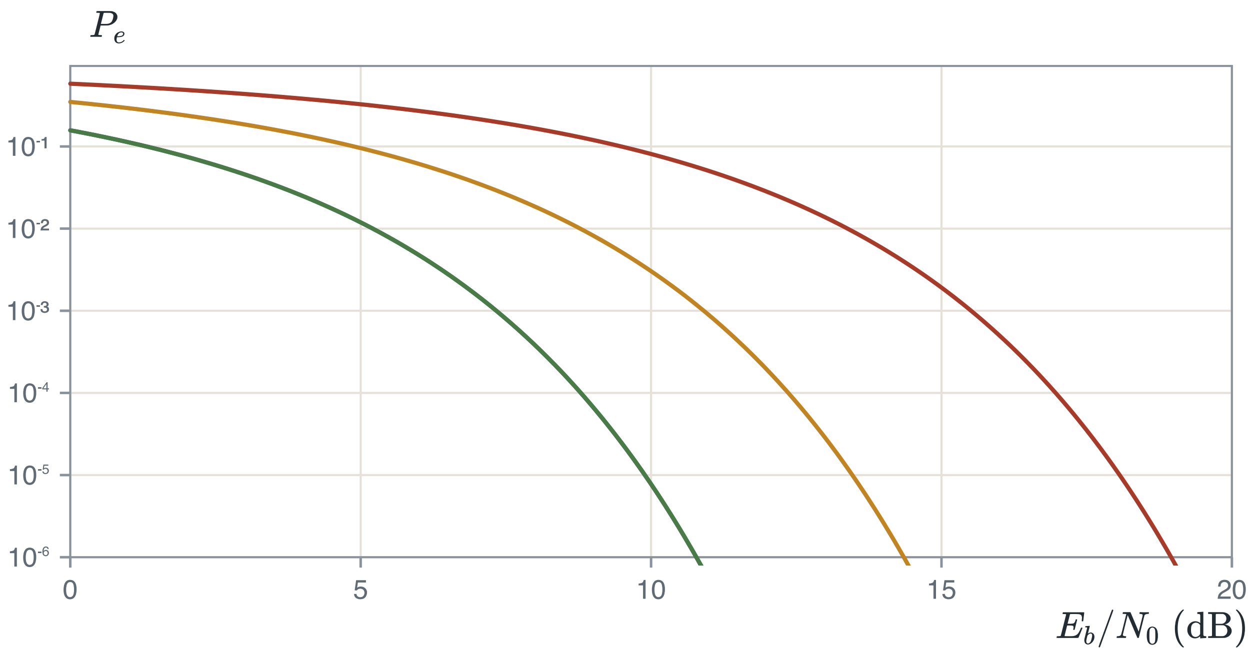
$$d_{jk} = \sqrt{2E_s} \text{ for every pair}, \qquad N_{\min} = M - 1$$

$$P_e \approx (M-1)\, Q\!\left(\sqrt{\frac{E_s}{N_0}}\right)$$

The distance does not depend on M at all. Adding waveforms costs nothing in distance — but each new waveform needs its own frequency slot, so the price is paid in bandwidth instead.

THE TRADE, STATED PLAINLY

PSK and QAM hold the bandwidth fixed as M grows and pay in energy, because the points crowd together. FSK holds the distance fixed and pays in bandwidth. A deep-space link, with bandwidth to spare and no power, chooses FSK; a mobile phone, with the opposite problem, chooses QAM. Neither is better in the abstract — it depends entirely on which resource is scarce.



Symbol error probability against energy per bit for four sizes of PSK: $M = 2, 4, 8, 16$ from left to right. $M = 2$ and $M = 4$ lie almost on top of each other, which is why QPSK is everywhere — it carries twice the bits of BPSK for the same energy per bit.

5.6 Summary

SCHEME	$d^2_{\min}$	$N_{\min}$	ANCHOR
BPSK	$4E_b$	1	PS CH8.6.1
BFSK, BASK	$2E_b$	1	PS CH9.5
M-PSK	$4E_s \sin^2(\pi/M)$	2	PS CH8.6.3
M-PAM	$12E_s/(M^2-1)$	$2(M-1)/M$	PS CH8.5.3
M-QAM	$6E_s/(M-1)$	3 at $M=16$	PS CH8.7.1
M-FSK	$2E_s$	$M-1$	PS CH9.5

Every row was obtained the same way: write the waveforms down, find the basis functions they need, place the points, measure the shortest distance, count how many pairs sit at it. The error probability then comes from chapter 4 without any further thought about waveforms. Convert once with $E_s = (\log_2 M)E_b$ when a question is stated per bit, and never convert twice.

What none of this says is how many bits a channel can carry at all. Every scheme here trades energy against bandwidth, and it is reasonable to ask whether there is a limit to the trade. Chapter 6 answers that, and the answer does not depend on which modulation is used.

CHAPTER 6

An introduction to information theory

Every chapter so far has taken the bits as given and worked on getting them across a channel. This one asks where they came from and how few of them a message really needs. The answer is one number, computed from the source alone, and it is both a measure of how much information the source produces and a limit on how few bits will carry it.

6.1 The information in one symbol

A **discrete memoryless source** emits one of K symbols each time, with fixed probabilities $p_1, \ldots, p_K$, chosen independently of what came before. How much is learnt when one arrives? If it was certain, nothing. If it was almost impossible, a great deal.

SELF-INFORMATION

$$I(s_k) = \log_a \frac{1}{p_k} = -\log_a p_k$$

The base sets the unit: $a=2$ gives **bits**, $a=e$ gives **nats**, $a=10$ gives **Hartleys**. This course uses base two throughout.

THREE PROPERTIES, AND WHAT FORCES THE LOGARITHM

$I(s_k) \geq 0$, because a probability is at most one and its logarithm at most zero.
$I(s_k) \geq I(s_j)$ **when** $p_k \leq p_j$: rarer means more informative.
$I(s_k s_j) = I(s_k) + I(s_j)$ **for independent symbols**, because their probabilities multiply and the logarithm turns a product into a sum. That third property is the one that rules out every function except a logarithm.

ONE HALVING IS ONE BIT

$-\log_2 \frac{1}{2} = 1$, so every halving of a probability adds exactly one bit. A symbol of probability $1/1000$ is about ten halvings away from certainty, so it carries about 10 bits. Most estimates in this chapter can be done that way, without a calculator.

6.2 Entropy

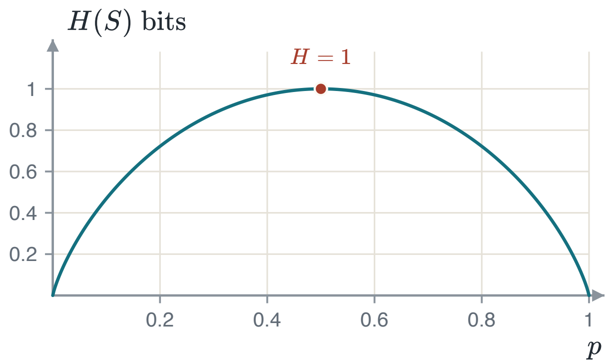
Self-information describes one symbol. Average it over the alphabet, weighting each symbol by how often it happens, and the result describes the source.

ENTROPY

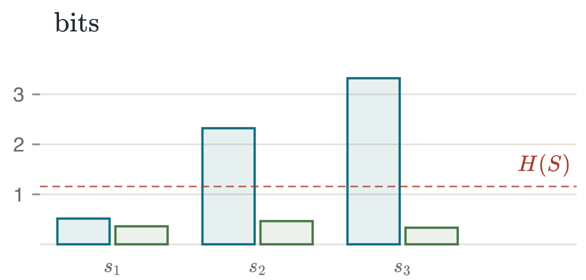
$$H(S) = E\big[I(s_k)\big] = -\sum_{k=1}^{K} p_k \log_2 p_k \quad \text{bits a symbol}$$

$$0 \leq H(S) \leq \log_2 K$$

Zero when one symbol has probability one — nothing is ever in doubt. At the top, $\log_2 K$, when all K are equally likely: putting $p_k = 1/K$ into the sum gives $\sum \frac{1}{K} \log_2 K = \log_2 K$.



The entropy of a binary source. One bit at $p = \frac{1}{2}$, nothing at either end, and flat near the top — a slightly unfair coin is almost as informative as a fair one.



The standard example. The left bar of each pair is what the symbol carries; the right bar is that weighted by how often it happens. The three right bars add to $H(S)$.

EXAMPLE 6.1 — THE ENTROPY OF A THREE-SYMBOL SOURCE

GIVEN $S = \{s_1, s_2, s_3\}$ with probabilities $0.7,\ 0.2,\ 0.1$.

FIND The entropy.

SOLUTION $H(S) = -0.7\log_2 0.7 - 0.2\log_2 0.2 - 0.1\log_2 0.1 = 0.3602 + 0.4644 + 0.3322 = 1.1568$ bits a symbol.

CHECK $\log_2 3 = 1.585$, and 1.1568 is below it, as it must be for a source that is not uniform. Numbering the three symbols would cost 2 bits each, so plain numbering wastes almost 0.85 bits every symbol. Recovering that is the whole of source coding.

Two numbers get confused here and should not be. $\log_2 K$ is the most a source of this size *could* carry. $\lceil \log_2 K \rceil$ is what a fixed-length code *costs*. The first gap is the source being uneven; the second is rounding.

6.3 Extended sources

Nothing forces a coder to work one symbol at a time. Group them in ns and treat each block as one symbol of a new alphabet. That is the **n-th extension**, written S^n, and it has K^n symbols.

EXTENSION

$$H(S^n) = n\,H(S)$$

The source is memoryless, so the symbols in a block are independent and their information adds. This is the third property of self-information, applied n times.

EXAMPLE 6.2 — THE SAME SOURCE, TWO AT A TIME

GIVEN The source of Example 6.1, extended by two.

FIND $H(S^2)$.

METHOD S^2 has $3^2 = 9$ symbols with probabilities $0.49, 0.14, 0.07, 0.14, 0.04, 0.02, 0.07, 0.02, 0.01$.

SOLUTION Summing $-p\log_2 p$ over all nine gives 2.3136 bits a block.

CHECK $2 \times 1.1568 = 2.3136$. The long way and the short way agree, and per symbol nothing has changed — 2.3136 over two symbols is 1.1568 each. What changes is how much of a whole bit gets wasted in rounding, and section 6.6 turns that into a bound.

ONLY BECAUSE IT IS MEMORYLESS

If the source had memory — as English does, where q is followed by u — the block probabilities would not be products, and $H(S^n)$ would be *less* than $nH(S)$. That difference is exactly the redundancy real compressors live on, and it is outside this course.

6.4 What a code costs

A source encoder turns each symbol into a string of bits, its **codeword**. The codewords need not all be the same length, and the good idea of this chapter is that they should not be: short ones for common symbols, long ones for rare.

AVERAGE LENGTH AND EFFICIENCY

$$\bar{L} = \sum_{k=1}^{K} p_k\, l_k, \qquad \eta = \frac{L_{\min}}{\bar{L}} = \frac{H(S)}{\bar{L}} \leq 1$$

The **source-coding theorem** says $\bar{L} \geq H(S)$ for any code from which the symbols can be recovered exactly. So $L_{\min} = H(S)$: the entropy is not merely a measure of information, it is a limit on how few bits will carry it.

SOURCE CODING IS NOT CHANNEL CODING

Source coding removes bits the message does not need. Channel coding adds bits back so that errors can be found and fixed. They pull in opposite directions, they are done in that order, and this course covers only the first.

Written English gives a concrete case. Its entropy is estimated at about 1.3 bits a letter, while a typical letter-by-letter variable-length code reaches $\bar{L} = 4.22$, so $\eta = 0.31$. Morse code is the same idea by hand: one dot for e, four symbols for z.

6.5 Codes that can be read back

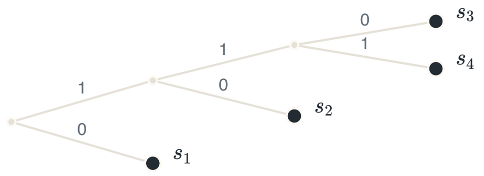
Short codewords are worth nothing if the receiver cannot tell where one ends. Two conditions matter, and they are not the same condition. A code is **uniquely decodable** if every string of bits it can produce comes from exactly one string of symbols. It is a **prefix code** if no codeword is the beginning of any other — which is stronger, and which lets the decoder name a symbol the moment its last bit arrives. A prefix code is also called **instantaneous**.

SYMBOL	CODE I	CODE II	CODE III
s_1	0	0	0
s_2	1	10	01
s_3	00	110	011
s_4	11	111	0111

Code I is not a prefix code and is not uniquely decodable either: the string 00 is both s_3 and $s_1 s_1$. Code II is a prefix code. Code III is not a prefix code, since 0 begins all three of the others.

CODE III IS THE INTERESTING ONE

It *can* be read back: every codeword begins with the only 0 and is then named by how many 1s follow, so any concatenation parses one way. What it cannot do is decode on the fly — after reading 011 the decoder still does not know whether it has s_3 or the start of s_4, and it must wait for the next 0. Uniquely decodable, but not instantaneous. The distinction is the whole point of the example.



Code II. Every symbol is at a leaf, so no path to one passes through another. That is what the prefix property looks like.

Code III. Every symbol sits on one path, each hanging off the node before it. Nothing is at a leaf but the last, and that is why the decoder must wait.

6.6 The Kraft inequality, and how close a code can get

Before choosing any codewords, ask whether the *lengths* can work. Think of the whole tree as one unit: a codeword of length l claims a branch and everything below it, a fraction 2^{-l} of the tree, and the claims cannot add to more than the tree there is.

KRAFT INEQUALITY

$$\sum_{k=1}^{K} 2^{-l_k} \leq 1$$

For the three codes above: 1.5, 1 and 0.9375. Code I fails, so no prefix code has its lengths. Code II uses the tree exactly. Code III passes with a sixteenth to spare — and is still not a prefix code, because the inequality says a prefix code with those *lengths* exists, not that these particular codewords are one. It is **necessary but not sufficient**.

HOW CLOSE A PREFIX CODE GETS

$$H(S) \leq \bar{L} < H(S) + 1$$

$$H(S) \leq \frac{L_n}{n} < H(S) + \frac{1}{n} \quad \Longrightarrow \quad \lim_{n\to\infty} \frac{L_n}{n} = H(S)$$

The ideal length for symbol k is $-\log_2 p_k$, almost never a whole number, so each length is rounded up and the rounding costs under one bit. Applying the same bound to the n-th extension spreads that one bit over n symbols.

WHEN THE BOUND IS TIGHT

If every probability is a power of two — a **dyadic** distribution, $p_k = 2^{-l_k}$ — nothing needs rounding. Then $\bar{L} = \sum l_k 2^{-l_k}$ and $H(S) = -\sum 2^{-l_k} \log_2 2^{-l_k} = \sum l_k 2^{-l_k}$: the same sum. So $\bar{L} = H(S)$ exactly and the code is perfect.

WHAT BLOCKING COSTS

The bound improves as $1/n$ and the alphabet grows as K^n. Reaching within 0.05 bits a symbol needs $n = 20$, and for a three-symbol source that is $3^{20} = 3.5 \times 10^9$ codewords. The theory says the entropy is reachable; the arithmetic says not this way, and both are true.

6.7 Huffman coding

The bound says a good code exists. Huffman coding builds one, and it builds the best one: no prefix code on single symbols has a smaller average length.

> THE ALGORITHM
>
> **1.** List the symbols in order of decreasing probability.
> **2.** Take the two least likely, label one 0 and the other 1, and merge them into one symbol whose probability is their sum. Re-sort.
> **3.** Repeat until two symbols remain, and label those 0 and 1.
> Each codeword is the labels along the path back to its symbol.

EXAMPLE 6.3 — THE STANDARD FIVE-SYMBOL SOURCE

GIVEN $p = 0.4,\ 0.2,\ 0.2,\ 0.1,\ 0.1.$

MERGING $0.1 + 0.1 = 0.2$; then the two smallest are 0.2 and 0.2, giving 0.4; then 0.2 and 0.4 give 0.6; then 0.4 and 0.6 finish it.

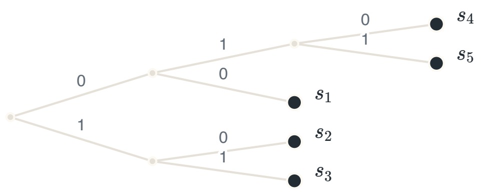
SOLUTION Lengths $2, 2, 2, 3, 3$, so $\bar{L} = 0.4(2) + 0.2(2) + 0.2(2) + 0.1(3) + 0.1(3) = 2.2$ bits a symbol.

EFFICIENCY $H(S) = 2.1219$ bits, so $\eta = 2.1219/2.2 = 0.9645$ — the code spends 3.68% more than the source strictly needs.

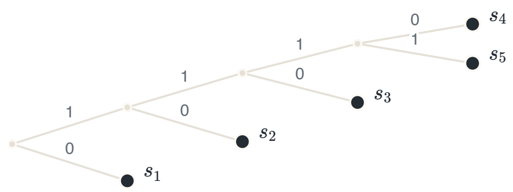
CHECK $2.1219 \le 2.2 < 3.1219$: the two-sided bound holds. A fixed-length code would have cost $\lceil \log_2 5 \rceil = 3$ bits, so Huffman saved 0.8 bits a symbol.

Two choices in the algorithm are free: which of a merged pair gets 0, and where a merged symbol goes when it ties with one already in the list. Different choices give different codes with the *same* average length — they must, because the minimum is unique even when the code is not. What separates them is how far the lengths spread about that average.

> VARIANCE OF THE CODEWORD LENGTH
>
> $$\sigma^2 = \sum_{k=1}^{K} p_k \left(l_k - \bar{L}\right)^2$$
>
> For the example: placing the merged symbol high gives lengths $2, 2, 2, 3, 3$ and $\sigma^2 = 0.16$; placing it low gives $1, 2, 3, 4, 4$, the same $\bar{L} = 2.2$, and $\sigma^2 = 1.36$.



Merged symbol placed high. Lengths $2, 2, 2, 3, 3$, average 2.2, variance 0.16. The tree is nearly balanced.



Merged symbol placed low. Lengths $1, 2, 3, 4, 4$, the same average 2.2, variance 1.36. The tree is a ladder.

WHY THE SMALLER VARIANCE IS WANTED

The encoder produces bits at a varying rate and the channel takes them at a fixed one, so a buffer sits between them. A code with wildly different lengths makes that buffer fill and empty unpredictably; a code with steady lengths does not. Both cost 2.2 bits on average, and only one is comfortable to build. **The rule:** on a tie, move the merged symbol as high as possible in the list.

6.8 Summary

RESULT	STATEMENT	ANCHOR
Self-information	$I(s_k) = -\log_2 p_k$	PS CH12.1.1
Entropy	$H(S) = -\sum_k p_k \log_2 p_k$	PS CH12.1.1
Bounds	$0 \le H(S) \le \log_2 K$	PS CH12.1.1
Extension	$H(S^n) = nH(S)$	PS CH12.1.1
Average length	$\bar{L} = \sum_k p_k l_k$, $\eta = H(S)/\bar{L}$	PS CH12.2
Source-coding theorem	$\bar{L} \ge H(S)$, so $L_{\text{min}} = H(S)$	PS CH12.2
Kraft	$\sum_k 2^{-l_k} \le 1$: necessary, not sufficient	PS CH12.3
Prefix bound	$H(S) \le \bar{L} < H(S) + 1$	PS CH12.2
Huffman	optimal, not unique; high placement gives least variance	PS CH12.3.1

Chapter 1 turned a waveform into bits. Chapters 2 to 5 got those bits across a channel and counted the errors. This chapter asked how few bits there needed to be in the first place, and answered with one number that the source itself decides. Everything between the two is engineering; the entropy is not.

APPENDIX A

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$$f_s > 2W \quad \text{(the sampling theorem, } W \text{ the highest frequency present)}$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \qquad |e| \le \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \qquad \mathrm{SQNR}\big|_{\mathrm{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n: at three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

RESULT	STATEMENT	CHAPTER
Companding	μ-law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = nf_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T - t), \qquad \left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0}$$

The filter matched to the pulse maximises the signal-to-noise ratio at the sampling instant, and the maximum depends on the energy of the pulse and not on its shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal)}, \qquad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1+\alpha}{2T}, \qquad 0 \le \alpha \le 1$$

The raised cosine meets the criterion for every roll-off α. At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly; at $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^{N} s_{ik}\psi_k(t), \qquad s_{ik} = \int_0^T s_i(t)\psi_k(t)\,dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^{N} s_{ik}^2, \qquad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg\min_i \Big\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \Big\} \quad \text{(MAP)}$$

$$\hat{s} = \arg\min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad \text{(ML, equal priors)}$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

ERROR PROBABILITY

$$P(\mathbf{s}_k \to \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le \frac{1}{M} \sum_k \sum_{j \ne k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right), \qquad P_e \approx N_{\min}\, Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first is exact for two points. The second is the union bound, and the third its nearest-neighbour form, where $N_{\min}$ is the *average* number of points at the minimum distance and need not be a whole number. The loose form $P_e \le (M-1)Q(\cdot)$ needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

SCHEME	$d^2_{\min}$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M-PSK	$4E_s \sin^2(\pi/M)$	2
M-PAM	$12E_s/(M^2-1)$	$2(M-1)/M$
M-QAM (square)	$6E_s/(M-1)$	3 at $M=16$, 3.5 at $M=64$
M-FSK	$2E_s$ for every pair	$M-1$

READING THE TABLE

$$P_e \approx N_{\min}\, Q\left(\sqrt{\frac{d^2_{\min}}{2N_0}}\right), \qquad E_s = (\log_2 M)E_b$$

Convert once between symbol and bit energy and never twice. BFSK and BASK need 3.01 dB more than BPSK for the same error probability; doubling M costs about 6 dB in PAM and about 5.33 dB going from 4-PSK to 8-PSK; and 16-QAM beats 16-PAM by 9.29 dB at the same energy.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \qquad H(S) = -\sum_{k=1}^{K} p_k \log_2 p_k$$

$$0 \le H(S) \le \log_2 K, \qquad H(S^n) = n\, H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^{K} p_k l_k, \qquad \eta = \frac{H(S)}{\bar{L}} \le 1$$

$$\sum_{k=1}^{K} 2^{-l_k} \le 1, \qquad H(S) \le \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n-th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \to H(S)$.

HUFFMAN CODES

$$\sigma^2 = \sum_{k=1}^{K} p_k \left(l_k - \bar{L}\right)^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.